



Graduation Project PCB Production Line

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A Project Graduation Document Submitted in Partial Fulfilment of the Requirements for
the BSc. Degree in Mechatronics Technology

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Project Abstract

The manufacturing of Printed Circuit Boards (PCBs) is a critical process in modern electronics production, requiring high precision, automation, and strict quality control. This graduation project presents the design and development of an automated PCB Production Line, integrating mechanical, electrical, control, and software systems to streamline the fabrication process. The project is structured into three core phases: system design, hardware implementation, and software development.

In the design phase, a comprehensive mechanical layout was developed to accommodate PCB handling, chemical processing, drilling, and inspection stages. The hardware phase involved the selection and integration of actuators, sensors, PLC-based controllers, and power distribution units to ensure reliable, safe, and repeatable operation. The software phase focused on developing real-time control algorithms, Human-Machine Interface (HMI) programming, and data logging systems to optimize throughput and minimize defect rates.

The integrated system was validated through simulated production cycles, demonstrating improved process efficiency, reduced manual intervention, and enhanced consistency compared to conventional manual methods. This project highlights the practical application of mechatronics principles in industrial automation and provides a scalable framework for small-to-medium electronics manufacturing. Future work will explore AI-based visual defect detection, IoT-enabled predictive maintenance, and full compliance with Industry 4.0 communication standards.

Keywords: PCB Manufacturing; Mechatronics Automation; Production Line Design; PLC Control; Industrial Monitoring; Quality Inspection

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Nomenclature

Nomenclature 1: Abbreviations and Acronyms

Term	Definition
AOI	Automated Optical Inspection
BOM	Bill of Materials
CAD	Computer-Aided Design
CAM	Computer-Aided Manufacturing
CNC	Computer Numerical Control
DC	Direct Current
EEPROM	Electrically Erasable Programmable Read-Only Memory
EMI	Electromagnetic Interference
FR-4	Flame Retardant 4 (Standard fiberglass-epoxy PCB substrate)
GND	Ground Reference
HMI	Human-Machine Interface
IoT	Internet of Things
MES	Manufacturing Execution System
PCB	Printed Circuit Board
PID	Proportional-Integral-Derivative Control
PLC	Programmable Logic Controller
PWM	Pulse Width Modulation
RPM	Revolutions Per Minute
RTOS	Real-Time Operating System
SMD	Surface Mount Device
SMT	Surface Mount Technology
SPI	Serial Peripheral Interface
SRAM	Static Random-Access Memory
THT	Through-Hole Technology
VCC	Positive Supply Voltage

Nomenclature 2: Symbols and Variables

Symbol	Definition	Unit
D	Drill bit diameter	mm
<i>F</i>	Feed rate (linear cutting speed)	mm/min
<i>I_{max}</i>	Maximum current rating of driver/motor	A
<i>N</i>	Spindle rotational speed	RPM
<i>P_{out}</i>	Mechanical output power	W
<i>R_{on}</i>	On-state resistance of motor driver	Ω
<i>S</i>	Step angle of stepper motor	$^{\circ}/\text{step}$
<i>T_{hold}</i>	Holding torque	N·m or kg·cm
<i>V_{cc}</i>	Supply voltage	V
θ	Angular position of servo/stepper	$^{\circ}$
μ	Coefficient of friction (belt/guide interface)	—
ω	Angular velocity	rad/s

Nomenclature 3:Units & Dimension

Unit	Description
mm	Millimetres
cm	Centimetres
m	Meters
g	Grams
kg	Kilograms
N	Newtons
V	Volts
A	Amperes
W	Watts
Hz	Hertz
RPM	Revolutions Per Minute
PPR	Pulses Per Revolution (encoder resolution)
ms	Milliseconds
μs	Microseconds
°C	Degrees Celsius
MPa	Megapascals (material stress/strength)

CHAPTER 1: Introduction and Aim of Work



Figure 1.1: Introduction and Aim of Work

1.1 INTRODUCTION

The electronics manufacturing industry has experienced unprecedented growth in recent decades, driven by the increasing demand for consumer electronics, communication devices, and embedded systems. At the heart of virtually every electronic device lies the Printed Circuit Board (PCB), which serves as the fundamental platform for mechanical support and electrical interconnection of electronic components. As technology advances and product life cycles shorten, the need for efficient, reliable, and cost-effective PCB manufacturing processes becomes increasingly critical.

Traditional PCB fabrication methods often rely on manual operations or semi-automated systems that can introduce inconsistencies, increase production time, and compromise quality control. These limitations are particularly pronounced in small to medium-scale production environments where full industrial automation may not be economically feasible. The integration of mechatronics principles—combining mechanical engineering, electronics, computer science, and control systems—offers a promising solution to bridge this gap by providing scalable, automated manufacturing solutions.

This graduation project presents the design and development of an automated PCB production line that integrates multiple manufacturing stages into a cohesive, streamlined system. The project leverages embedded control systems based on Arduino platforms, precision electromechanical mechanisms including servo motors and CNC systems, and comprehensive sensor integration to achieve consistent, high-quality PCB fabrication suitable for educational and small-scale production environments.

1.2 PROBLEM STATEMENT

The current landscape of PCB manufacturing, particularly in educational institutions and small-scale production facilities, faces several significant challenges:

1.2.1 Manual Handling Issues: Traditional PCB fabrication processes involve extensive manual handling between different production stages, leading to increased defect rates, production bottlenecks, and worker fatigue.

1.2.2 Lack of Integration: Disconnected manufacturing processes for drilling, routing, marking, and assembly preparation result in poor workflow efficiency, increased setup times, and difficulty maintaining consistent quality standards.

1.2.3 Quality Control Challenges: Manual or semi-automated processes are prone to human error, resulting in variations in hole positioning accuracy, inconsistent marking quality, and misalignment during component assembly.

1.2.4 Limited Scalability: Conventional setups lack the flexibility to adapt to different PCB sizes, designs, or production volumes without significant reconfiguration and downtime.

1.2.5 High Operational Costs: Reliance on manual labor, rework due to defects, and inefficient material handling contribute to elevated production costs and reduced competitiveness.

These challenges necessitate the development of an integrated, automated production line that can address these issues through systematic mechatronics design, embedded control, and process optimization.

1.3 PROJECT OBJECTIVES

The primary objectives of this graduation project are:

1.3.1 Design and implement an integrated PCB production line that automates sequential manufacturing processes from raw substrate to assembly-ready boards.

1.3.2 Develop a reliable PCB feeding and transport mechanism using conveyor belt systems with precise positioning control to ensure consistent board spacing and alignment.

1.3.3 Integrate CNC drilling and routing stations capable of creating precise holes for component leads and vias, as well as performing board profiling and outline cutting operations.

1.3.4 Design and implement an automated PCB flipping mechanism to enable double-sided processing without manual intervention, ensuring accurate registration between front and back operations.

1.3.5 Develop a CNC-based marking system using pen plotting technology to mark component positions, reference designators, polarity indicators, and assembly guidance information on PCB surfaces.

1.3.6 Create a structured assembly station that utilizes marked positions to facilitate accurate component placement and orientation during manual or semi-automated assembly processes.

1.3.7 Implement embedded control systems using Arduino platforms to coordinate all production stages, manage motion control, process sensor feedback, and ensure synchronized operation.

1.3.8 Ensure system reliability, safety, and ease of operation through comprehensive testing, validation, and the implementation of error

1.3.9 Design and implement an automated spraying system for uniform insulation layer application with controlled coating thickness.

1.4 PROJECT SCOPE

This project encompasses the complete design, development, and testing of a PCB production line prototype suitable for educational demonstration and small-scale production. The scope is defined across multiple engineering domains:

Mechanical Design Scope:

- Conveyor belt transport system with adjustable speed control
- CNC drilling and routing machine with three-axis motion control
- PCB flipping mechanism utilizing servo-driven rotation
- CNC pen plotter for component marking and labeling
- Assembly station with ergonomic design and component organization
- Structural frames and support systems using aluminum profiles and acrylic components
- Dedicated spraying chamber with precision nozzle system
- Drying/curing zone for insulation layer solidification
- Fluid control system for ink flow regulation

Electrical and Control Systems Scope:

- Arduino-based embedded control architecture (Arduino Uno/Mega)
- Servo motor systems for precise angular positioning
- Stepper motor drivers for CNC axis control
- Limit switches and proximity sensors for position detection
- Emergency stop circuits and safety interlocks
- Power distribution units with voltage regulation

Software Development Scope:

- Firmware development using C/C++ for Arduino platforms
- G-code generation and interpretation for CNC operations
- Sequential process control algorithms with state machine implementation
- Real-time sensor data processing and feedback control
- Error detection and recovery protocols
- Basic Human-Machine Interface (HMI) for system operation

Exclusions from Scope:

- Solder paste application and reflow soldering processes
- Automated optical inspection (AOI) with machine vision
- Chemical etching processes for PCB fabrication
- Surface mount technology (SMT) pick-and-place automation
- Industrial PLC-based control systems (future enhancement)

1.5 SYSTEM OVERVIEW

The PCB production line operates through five sequential, integrated stages that transform raw PCB substrates into assembly-ready boards:

Stage 1 – Insulation Layer Application (Spraying):

Raw PCB substrates are introduced into a dedicated spraying chamber where a specialized insulating ink is uniformly applied to the board surface. The chamber utilizes automated spray nozzles controlled by precision fluid pumps and coordinated motion systems to ensure consistent coating thickness and complete coverage. Following application, the boards pass through a controlled drying/curing zone to solidify the insulation layer. This protective coating enhances electrical isolation, prevents short circuits during subsequent machining operations, and improves the mechanical durability of the final product.

Stage 2 – CNC Drilling and Routing:

A CNC-controlled drilling station equipped with a high-speed spindle creates precise holes for through-hole component leads and interlayer vias. The same CNC system performs routing operations to cut the PCB to final dimensions, create board outlines, and mill slots or cutouts as required by the design. Positional accuracy is maintained through stepper motor control and limit switch homing.

Stage 3 – PCB Flipping:

An automated flipping mechanism engages the processed PCB and rotates it 180 degrees to enable operations on the opposite side. The mechanism utilizes servo motors for controlled rotation and vacuum or mechanical grippers for secure board handling. This stage eliminates the need for manual board reversal, reducing handling time and potential damage.

Stage 4 – Component Marking:

A CNC pen plotter marks component positions, reference designators (R1, C5, U3, etc.), polarity indicators, and other assembly guidance information directly onto the PCB surface. The plotting system uses standard drawing pens or markers and follows vector paths generated from the PCB design data, providing clear visual references for assembly technicians.

Stage 5 – Assembly Station:

The final station provides a structured workspace for component placement. The marked positions guide operators in placing components with correct orientation and location. The station may include component organizers, magnification aids, and ergonomic seating to facilitate efficient manual assembly

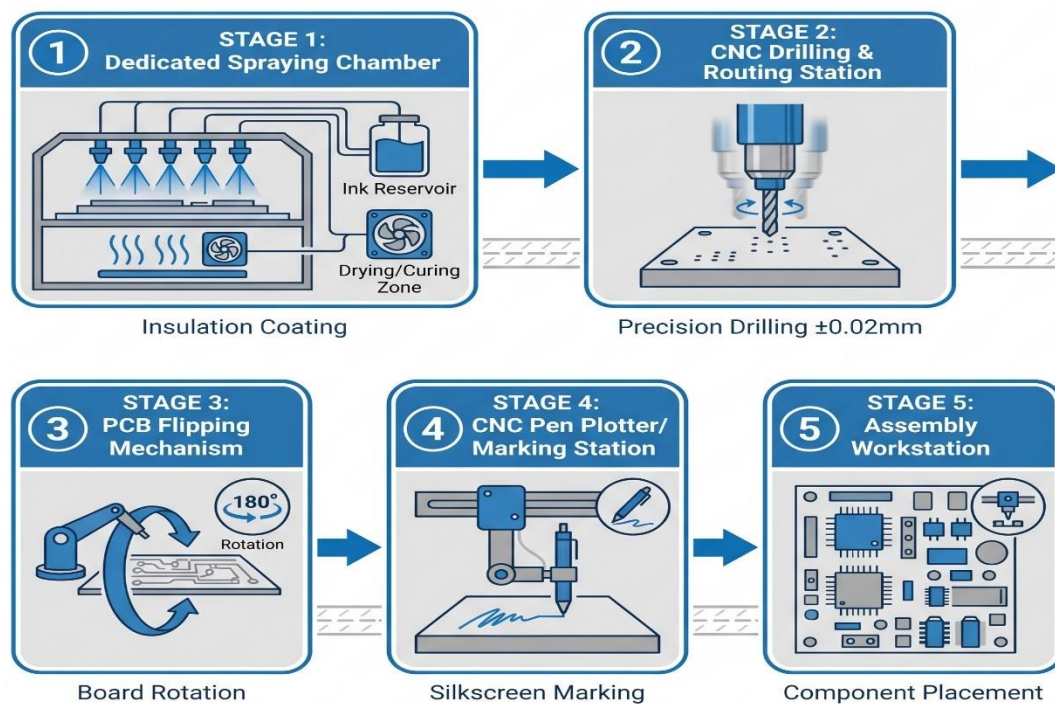


Figure 1.2:SYSTEM OVERVIEW

1.6 REPORT ORGANIZATION

This report is organized into five comprehensive chapters that document all aspects of the graduation project:

Chapter 1: Introduction and Aim of Work

Presents the project background, problem statement, objectives, scope, system overview, and report organization. This chapter establishes the context and motivation for the project.

Chapter 2: Literature Review and Theoretical Background

Provides comprehensive coverage of PCB manufacturing processes, CNC technology fundamentals, embedded control systems, servo and stepper motor theory, sensor technologies, and automation principles. This chapter also reviews relevant previous work and industry standards.

Chapter 3: System Design and Methodology

Details the complete system design including mechanical design calculations and CAD drawings, electrical system architecture and circuit diagrams, control system development with I/O mapping, software architecture with flowcharts and algorithms, and component selection criteria with technical specifications.

Chapter 4: Implementation and Testing

Describes the hardware implementation process including mechanical assembly and electrical wiring, software implementation with code examples, system integration procedures, testing methodologies for individual subsystems and the complete line, results analysis with performance metrics, and troubleshooting activities.

Chapter 5: Conclusion and Future Work

Summarizes the project achievements and lessons learned, draws conclusions about system performance and objectives attainment, identifies limitations and challenges encountered, and recommends potential improvements and future developments including PLC migration, machine vision integration, and Industry 4.0 connectivity

1.7 EXPECTED OUTCOMES

1.7.1 Upon successful completion of this project, the following outcomes are expected:

1.7.2 A fully functional prototype of an integrated PCB production line capable of automated drilling, routing, flipping, marking, and assembly preparation.

1.7.3 Demonstrated automation of PCB handling and processing with minimal manual intervention between stages.

1.7.4 Improved production efficiency and consistency compared to conventional manual methods, with measurable reductions in processing time and defect rates.

1.7.5 Enhanced positional accuracy in drilling and marking operations through CNC control and embedded feedback systems.

1.7.6 A scalable and modular system architecture that can be adapted for various PCB sizes, designs, and production volumes.

1.7.7 Comprehensive documentation including design specifications, assembly instructions, operating procedures, and maintenance guidelines.

1.7.8 Validation of the feasibility of low-cost embedded automation for educational and small-scale PCB manufacturing applications.

1.7.9 Uniform insulation coating with controlled thickness

1.7.10 Enhanced electrical isolation and short-circuit prevention

The project aims to contribute to the advancement of automated manufacturing systems in the electronics industry and provide a practical framework for implementing mechatronics solutions in real-world production environments. By demonstrating the integration of Arduino-based control, precision mechanical systems, and automated process sequencing, this project serves as a foundation for future enhancements toward industrial-grade PCB manufacturing automation.

1.8 PROJECT TIMELINE AND MILESTONES

Table 1-1: Time Line & Milestones

Phase	Activity	Duration	Deliverable
Phase 1	Literature Review & Requirements Analysis	3 weeks	Requirements Document
Phase 2	Mechanical Design & CAD Modeling	4 weeks	3D Models & Drawings
Phase 3	Electrical System Design	3 weeks	Circuit Diagrams & BOM
Phase 4	Control System Development	4 weeks	Firmware & Control Logic
Phase 5	Mechanical Assembly	3 weeks	Assembled Prototype
Phase 6	Electrical Integration & Wiring	2 weeks	Integrated System
Phase 7	Testing & Validation	3 weeks	Test Reports & Results
Phase 8	Documentation & Report Writing	3 weeks	Final Project Report

CHAPTER 2: Review and Theoretical Background

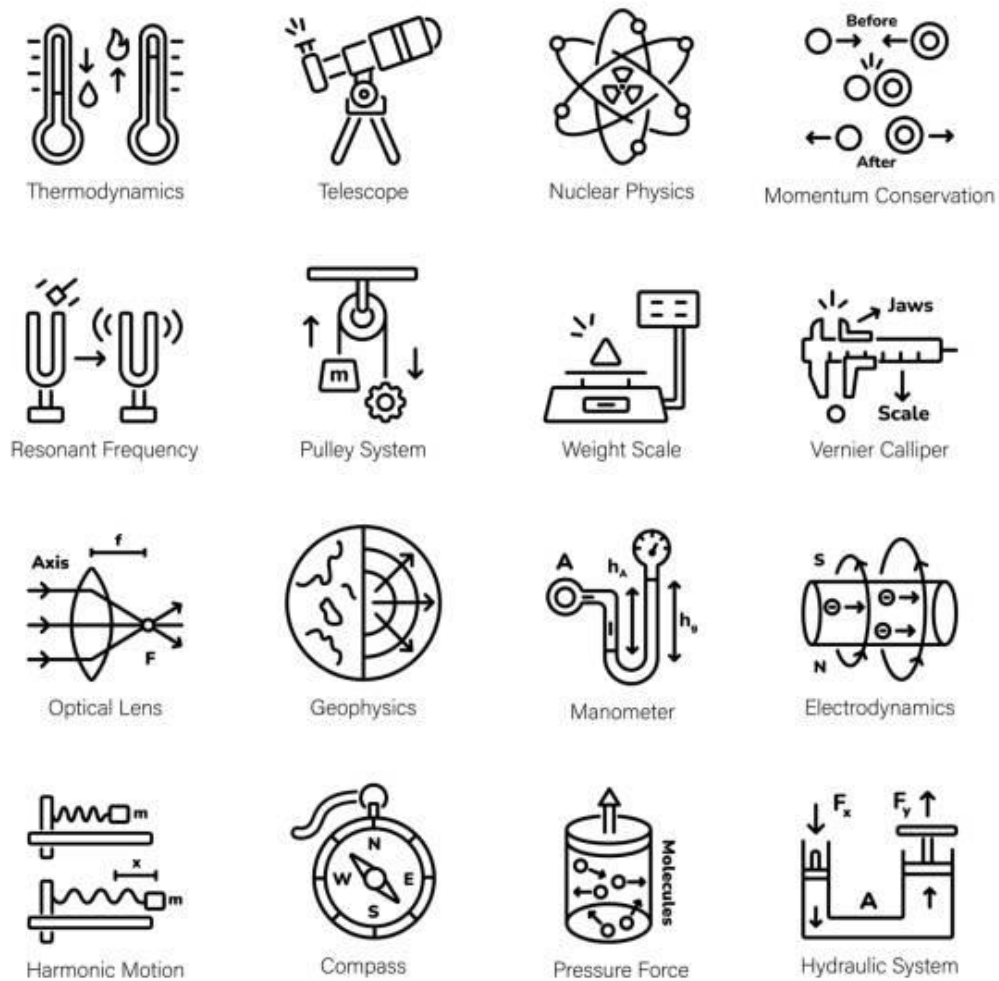


Figure 2.1: Review and Theoretical Background

2.1 INTRODUCTION

This chapter provides a comprehensive review of the theoretical foundations and previous work relevant to automated PCB manufacturing systems. The chapter covers fundamental concepts of PCB fabrication processes, insulation coating techniques, automated spraying systems, CNC technology principles, embedded control systems, electromechanical actuators, and sensor technologies. Additionally, this chapter examines previous research and industrial implementations in the field of automated electronics manufacturing, establishing the theoretical framework upon which this project is built.

2.2 PRINTED CIRCUIT BOARD MANUFACTURING FUNDAMENTALS

2.2.1 PCB Structure and Components

A Printed Circuit Board (PCB) is a multilayered structure that provides mechanical support and electrical connections for electronic components. The basic PCB consists of a non-conductive substrate material, typically fiberglass-reinforced epoxy (FR-4), with conductive copper traces etched or printed on one or both sides. Modern PCBs can have multiple layers (up to 12 or more) interconnected through plated through-holes called vias.

The primary functions of a PCB include:

Mechanical support for electronic components

Electrical interconnection between components through conductive traces

Heat dissipation from components

Signal integrity maintenance through controlled impedance traces

Electromagnetic interference (EMI) shielding in multilayer designs

2.2.2 PCB Fabrication Process Steps

2.2.2.1 The standard PCB manufacturing process involves several critical stages:

2.2.2.2 Design and Layout: PCB design is created using Computer-Aided Design (CAD) software, generating Gerber files that define copper layers, drill data, and solder mask patterns.

2.2.2.3 Material Preparation: Copper-clad laminate sheets are cut to the required dimensions and cleaned to remove contaminants.

2.2.2.4 Insulation Coating Application: A protective insulating layer is applied to the PCB surface using spray coating, screen printing, or lamination techniques. This layer prevents short circuits and provides electrical isolation.

2.2.2.5 Drilling: Precision holes are drilled for through-hole components and vias using CNC-controlled drilling machines with tungsten carbide drill bits. Typical drill diameters range from 0.2 mm to 6.0 mm with positional accuracy of ± 0.025 mm.

2.2.2.6 Plating and Etching: For multilayer boards, holes are plated with copper to create electrical connections between layers. Unwanted copper is removed through chemical etching processes.

2.2.2.7 Solder Mask Application: A protective polymer layer is applied to prevent oxidation and solder bridges during assembly.

2.2.2.8 Silkscreen Printing: Component designators, logos, and assembly information are printed on the board surface.

2.2.2.9 Surface Finish: A protective coating (HASL, ENIG, or OSP) is applied to exposed copper pads to ensure solderability.

2.2.2.10 Electrical Testing: Boards undergo continuity testing and insulation resistance verification.

2.2.3 Insulation and Coating Techniques

Insulation coating is a critical process in PCB manufacturing that ensures electrical isolation between conductive traces and prevents short circuits. Common coating methods include:

Spray Coating:

- Utilizes atomized paint/ink particles propelled by compressed air
- Provides uniform coverage over complex geometries
- Suitable for automated production lines
- Typical coating thickness: 25-100 μm
- Requires controlled environment (temperature, humidity, ventilation)

Screen Printing:

- Uses mesh screens to transfer ink onto PCB surface
- Cost-effective for high-volume production
- Limited to flat surfaces
- Coating thickness: 15-50 μm

Lamination:

- Applies pre-cured insulating films under heat and pressure
- Excellent uniformity and consistency
- Higher material cost
- Used primarily for multilayer PCBs

Key Parameters for Insulation Coating:

- Viscosity: 200-1000 cP (centipoise)
- Drying temperature: 80-150°C
- Curing time: 15-60 minutes
- Dielectric strength: >20 kV/mm
- Adhesion strength: >1.5 N/mm

2.3 AUTOMATED SPRAYING SYSTEMS

2.3.1 Spray Coating Principles

Automated spray coating systems utilize fluid dynamics and atomization principles to apply uniform layers of material onto substrates. The process involves:

Atomization:

Liquid coating is broken into fine droplets (10-200 μm diameter)

Achieved through pneumatic pressure, hydraulic pressure, or electrostatic forces

Droplet size affects coating quality and material efficiency

Spray Pattern Formation:

Nozzle geometry determines spray cone angle and distribution

Common patterns: circular, fan, conical

Overlapping passes ensure uniform coverage

Transfer Efficiency:

Ratio of coating material deposited on substrate vs. total material used

Typical efficiency: 40-70% for conventional spray

Improved to 80-95% with electrostatic spray

2.3.2 Nozzle Types and Selection

Spray nozzles are critical components that determine coating quality and efficiency. Common types include:

Air Spray Nozzles:

Use compressed air to atomize coating material

Provide fine finish and good control

Operating pressure: 2-6 bar

Suitable for low to medium viscosity materials (100-500 cP)

Airless Spray Nozzles:

Rely on high hydraulic pressure for atomization

Higher transfer efficiency

Operating pressure: 100-300 bar

Suitable for high-viscosity coatings

HVLP (High Volume Low Pressure) Nozzles:

Use high air volume at low pressure (0.7-1.0 bar)

Transfer efficiency: 65-85%

Reduced overspray and material waste

Ideal for precision coating applications

Nozzle Selection Criteria:

Orifice size: 0.5-2.0 mm (determines flow rate)

Spray angle: 15°-120° (determines coverage width)

Material compatibility (stainless steel, brass, ceramic)

Flow rate: 50-500 mL/min

2.3.3 Fluid Dynamics in Spray Systems

Understanding fluid dynamics is essential for optimizing spray coating performance:

Flow Rate Control:

Governed by Bernoulli's equation and orifice flow principles

$$Q = C_d \times A \times \sqrt{(2\Delta P/\rho)}$$

Where:

Q = Flow rate (m^3/s)

C_d = Discharge coefficient (0.6-0.98)

A = Orifice area (m^2)

ΔP = Pressure differential (Pa)

ρ = Fluid density (kg/m^3)

Viscosity Effects:

Higher viscosity requires higher pressure for atomization

Temperature control critical for consistent viscosity

Typical operating temperature: 20-30°C

Droplet Size Distribution:

Affected by nozzle design, pressure, and fluid properties

Smaller droplets provide smoother finish but increase overspray

Sauter Mean Diameter (SMD): 20-100 μm typical

2.3.4 Drying and Curing Processes

After spray application, the coating must be dried and cured to achieve desired properties:

Drying Mechanisms:

Evaporation: Solvent removal through heating (80-120°C)

Oxidation: Reaction with atmospheric oxygen

Coalescence: Polymer particle fusion

Curing Methods:

Thermal Curing: Heat-activated crosslinking (120-180°C, 15-60 min)

UV Curing: Ultraviolet radiation initiates polymerization (seconds)

Two-Component Curing: Chemical reaction between resin and hardener

Curing Zone Design:

Conveyor speed: 0.5-2.0 m/min

Temperature zones: Preheat (60°C), Cure (150°C), Cool (40°C)

Residence time: 20-40 minutes total

Air circulation: Forced convection for uniform heating

2.4 COMPUTER NUMERICAL CONTROL (CNC) TECHNOLOGY

2.4.1 CNC System Architecture

Computer Numerical Control (CNC) systems are automated machine tools that execute pre-programmed sequences of commands to control machine movements with high precision. A typical CNC system consists of:

Controller Unit:

- Processes G-code instructions and generates motion commands
- Performs interpolation calculations for smooth trajectory generation
- Manages input/output signals for sensors and actuators
- Provides user interface for program editing and machine control

Drive System:

- Amplifies control signals to power motors
- Converts low-power commands into high-power motor drive signals
- Provides current limiting and overload protection
- Supports various motor types (stepper, servo, DC)

Feedback System:

- Provides real-time position and velocity information
- Uses encoders, resolvers, or linear scales
- Enables closed-loop control for improved accuracy
- Typical encoder resolution: 1000-5000 pulses/revolution

Machine Tool:

- The mechanical structure performing the work (spindle, axes)
- Provides rigidity and vibration damping
- Incorporates linear guides and ball screws for precision motion
- Typical positioning accuracy: ± 0.01 - 0.05 mm

Human-Machine Interface (HMI):

- Allows operator input and monitoring
- Displays machine status, alarms, and program information
- Provides manual control modes (jog, handwheel)
- Modern systems use touchscreen panels

2.4.2 G-Code Programming

G-code (RS-274) is the standard programming language for CNC machines. Common G-code commands used in PCB manufacturing include:

Motion Commands:

- **G00:** Rapid positioning (non-cutting movement)
- **G01:** Linear interpolation (cutting movement at specified feed rate)
- **G02/G03:** Circular interpolation (clockwise/counterclockwise arcs)
- **G04:** Dwell (pause for specified time)

Drilling Cycles:

- **G81:** Standard drill cycle (automated drilling operation)
- **G82:** Drill cycle with dwell (for counterboring)
- **G83:** Peck drilling cycle (for deep holes)
- **G80:** Cancel canned cycle

Spindle and Coolant Control:

- **M03:** Spindle on clockwise
- **M04:** Spindle on counterclockwise
- **M05:** Spindle stop
- **M08:** Coolant on
- **M09:** Coolant off

Program Control:

- **M00:** Program stop
- **M01:** Optional stop
- **M02:** Program end
- **M30:** Program end and rewind

Example PCB Drilling G-code:

```
1  %
2  O1000 (PCB Drilling Program)
3  G21 (Set units to millimeters)
4  G90 (Absolute positioning)
5  G17 (Select XY plane)
6  G80 (Cancel any active cycles)
7
8  (Tool 1: 0.8mm Drill Bit)
9  T01 M06
10 M03 S12000 (Spindle on, 12000 RPM)
11 G81 R2.0 Z-1.5 F100 (Drill cycle, retract 2mm, depth 1.5mm, feed 100mm/min)
12
13 (Drill holes)
14 X10.5 Y15.2
15 X25.3 Y15.2
16 X10.5 Y28.7
17 X25.3 Y28.7
18
19 G80 (Cancel drill cycle)
20 M05 (Spindle stop)
21 M30 (Program end)
22 %
```

Figure 2.2:Example PCB Drilling G-code

2.4.3 CNC Spindle Systems

PCB drilling requires high-speed spindles operating at 12,000-60,000 RPM to achieve clean holes in fiberglass-epoxy composites. Key spindle parameters include :

Speed and Power:

- Speed Range: 12,000-60,000 RPM for PCB applications
- Power Rating: 0.5-3.0 kW depending on application
- Torque: 0.05-0.5 N·m (high speed, low torque)

Accuracy Specifications:

- Runout Accuracy: < 0.005 mm for precision drilling
- Axial stiffness: >50 N/μm
- Radial stiffness: >30 N/μm

Cooling Systems:

- **Air-cooled:** Simpler, lower cost, suitable for intermittent operation
- **Water-cooled:** Better temperature stability, continuous operation capability
- Typical operating temperature: 20-40°C

Tool Holding:

- Collet systems (ER11, ER16, ER20) for drill bits 0.2-6.0 mm
- Clamping force: 500-2000 N
- Tool change time: 2-10 seconds (automatic), 30-60 seconds (manual)

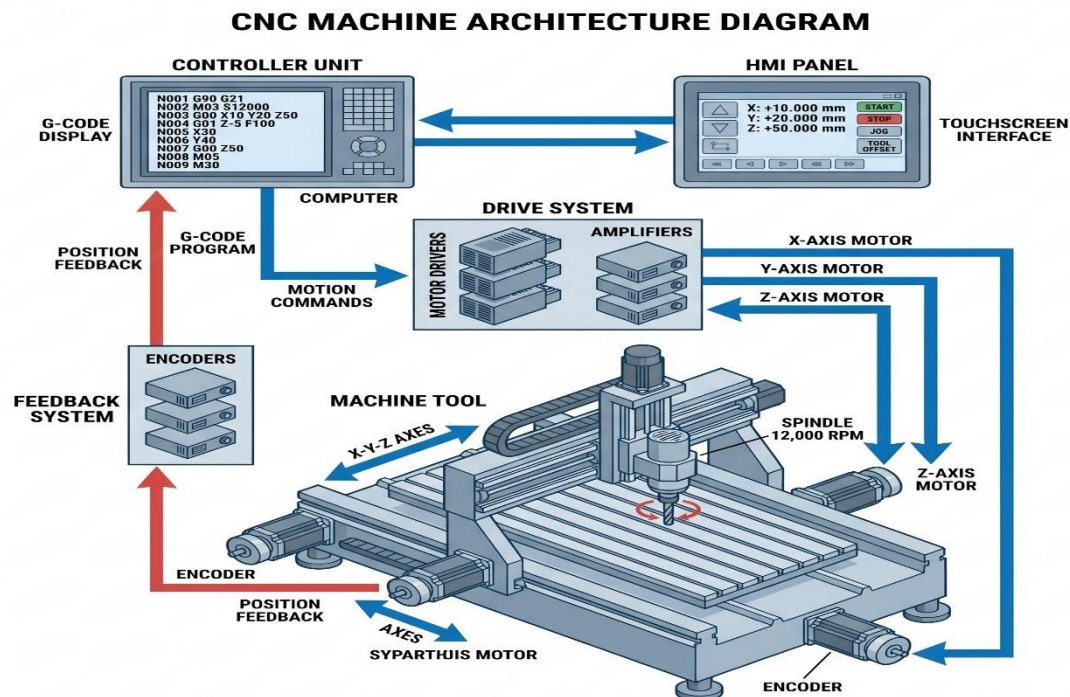


Figure 2.6:CNC ARCHITECTURE Diagram

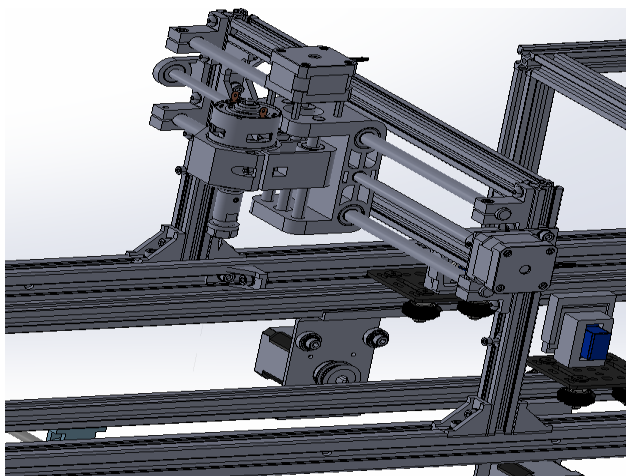


Figure 2.4:CNC Real Project Design 1

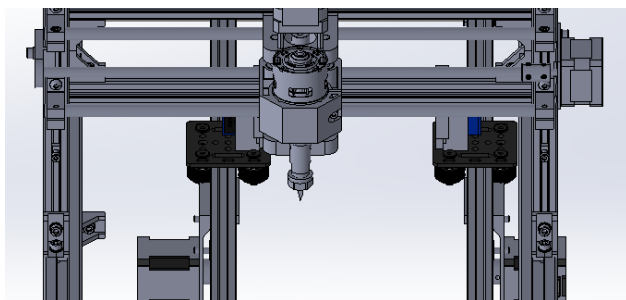


Figure 2.5:CNC Real Project Design 2

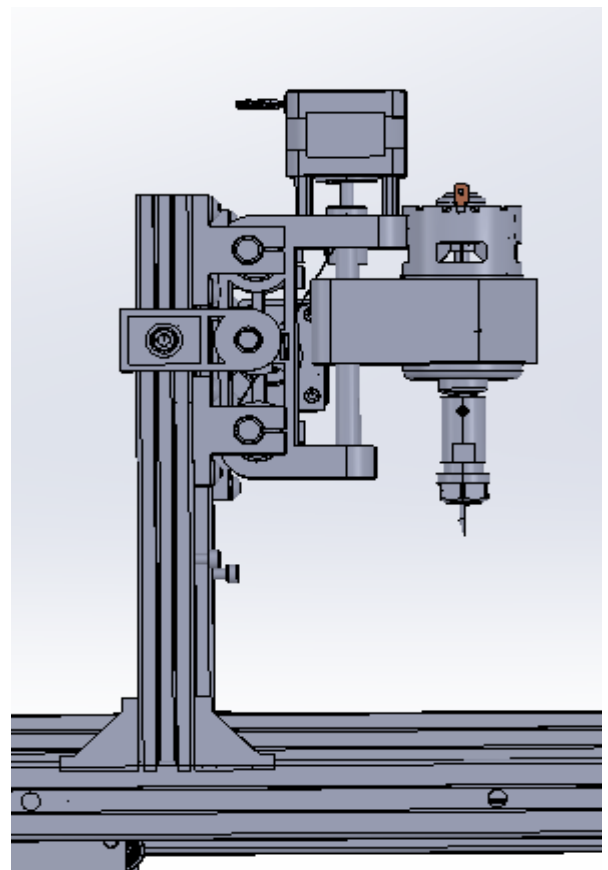


Figure 2.3:CNC Real Project Design 3

2.5 EMBEDDED CONTROL SYSTEMS

2.5.1 Arduino Platform Architecture

The Arduino platform is open-source electronics prototyping platform based on AVR microcontrollers, widely used in mechatronics and automation projects due to its simplicity, extensive library support, and cost-effectiveness.

Arduino Uno Specifications:

- **Microcontroller:** ATmega328P (8-bit AVR)
- **Operating Voltage:** 5V DC
- **Digital I/O Pins:** 14 (6 PWM outputs)
- **Analog Input Pins:** 6 (10-bit resolution)
- **Clock Speed:** 16 MHz
- **Flash Memory:** 32 KB (0.5 KB used by bootloader)
- **SRAM:** 2 KB
- **EEPROM:** 1 KB
- **Input Current per Pin:** 20-40 mA (absolute max)

Arduino Mega 2560 Specifications (for complex projects):

- **Microcontroller:** ATmega2560
- **Digital I/O Pins:** 54 (15 PWM outputs)
- **Analog Input Pins:** 16 (10-bit resolution)
- **Flash Memory:** 256 KB (8 KB used by bootloader)
- **SRAM:** 8 KB
- **EEPROM:** 4 KB

Communication Interfaces:

- **UART (Serial):** 9600-115200 bps for PC communication
- **I2C:** 100-400 kHz for sensor/actuator communication
- **SPI:** Up to 8 MHz for high-speed data transfer
- **USB:** Virtual COM port for programming and debugging



Figure 2.8:Arduino Uno Board

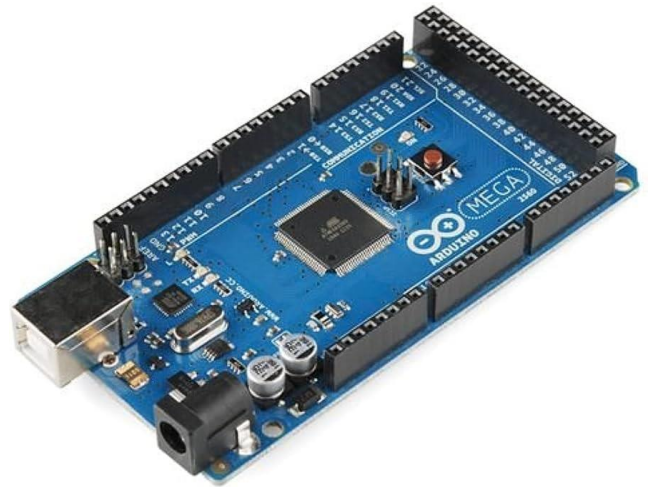


Figure 2.7:Arduino mega Board

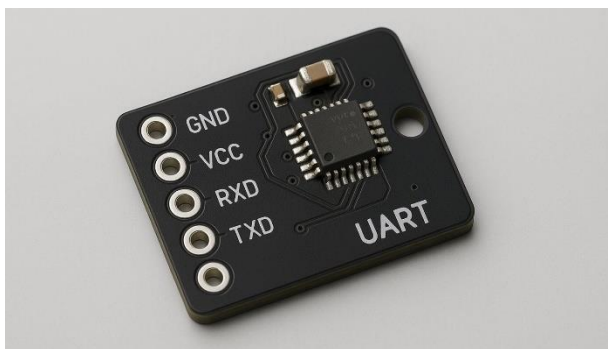


Figure 2.10:UART Protocol

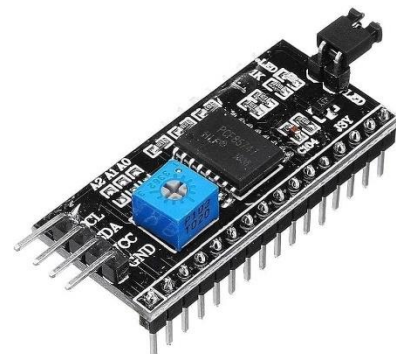


Figure 2.9:I2C Protocol

2.5.2 Real-Time Control Considerations

Embedded control systems for production lines must meet real-time constraints to ensure synchronized operation. Key considerations include:

Deterministic Timing:

- Critical operations must complete within guaranteed time bounds
- Arduino loop time: typically, 1-10 ms for simple programs
- Interrupt latency: 3-5 clock cycles (0.1875-0.3125 μ s at 16 MHz)
- Timer resolution: 62.5 ns (16 MHz clock)

Interrupt Handling:

- External events (limit switches, emergency stops) require immediate response
- Arduino supports 2 external interrupts (Uno) or 6 (Mega)
- Pin change interrupts available on all I/O pins
- Timer interrupts for periodic tasks

Task Scheduling:

- Multiple processes (conveyor motion, CNC operations, sensor monitoring) must be coordinated
- **Cooperative multitasking:** Tasks yield control voluntarily
- **Preemptive multitasking:** RTOS switches tasks based on priority
- Common Arduino scheduling methods:
 - `millis()`-based non-blocking timing
 - Timer interrupts for precise timing
 - State machines for sequential processes

Watchdog Timers:

- System recovery from software faults
- Resets microcontroller if program hangs
- Arduino watchdog timeout: 16 ms - 8 seconds
- Essential for unattended industrial operation

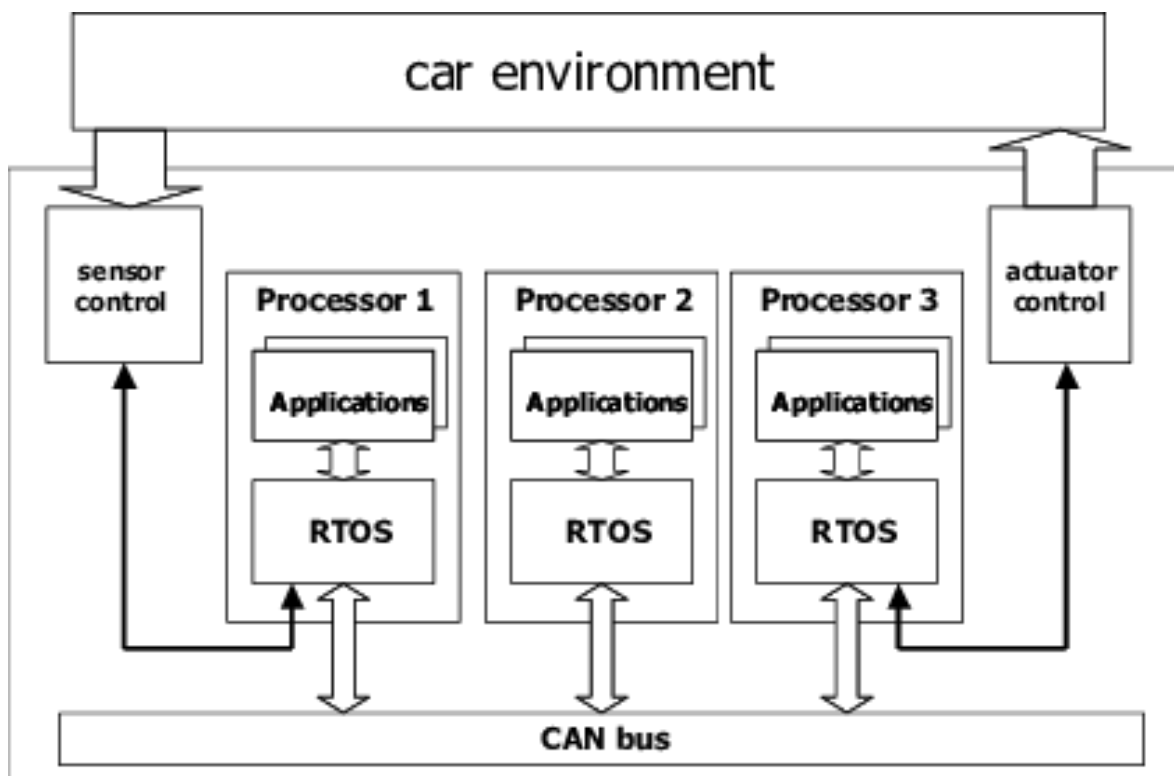


Figure 2.11:Real Time Control System

2.5.3 Firmware Development for PCB Production Line

The control firmware for the PCB production line implements several critical functions:

Conveyor Control:

```
1 // Conveyor speed control using PWM
2 const int conveyorPWM = 9;
3 int targetSpeed = 128; // 0-255 (50% duty cycle)
4
5 void controlConveyor(int speed) {
6   analogWrite(conveyorPWM, speed);
7 }
8
9 // Board detection using IR sensor
10 const int boardSensor = 2;
11 volatile bool boardDetected = false;
12
13 void sensorISR() {
14   boardDetected = true;
15 }
16
17 void setup() {
18   attachInterrupt(digitalPinToInterrupt(boardSensor), sensorISR, RISING);
19 }
```

Figure 2.12:Conveyor Control

CNC Motion Control:

```
1 // Stepper motor control for CNC axes
2 #include <AccelStepper.h>
3
4 AccelStepper xAxis(1, 2, 3); // step, dir pins
5 AccelStepper yAxis(1, 4, 5);
6
7 void moveToPosition(float x, float y) {
8   xAxis.moveTo(x * stepsPerMm);
9   yAxis.moveTo(y * stepsPerMm);
10
11   while (xAxis.distanceToGo() != 0 || yAxis.distanceToGo() != 0) {
12     xAxis.run();
13     yAxis.run();
14   }
15 }
```

Figure 2.13:CNC Motion Control

Servo Control for Flipping:

```
1  #include <Servo.h>
2
3  Servo flipServo;
4  const int flipAngle = 180;
5  const unsigned long flipDelay = 2000; // ms
6
7  void flipPCB() {
8      flipServo.write(0);
9      delay(500); // Wait for gripper engagement
10     flipServo.write(flipAngle);
11     delay(flipDelay);
12     flipServo.write(0);
13 }
```

Figure 2.14: Servo Control for Flipping

2.6 SENSOR TECHNOLOGIES

2.6.1 Limit Switches and End Stops

Mechanical limit switches provide reliable position detection for CNC axis homing and overtravel protection. Common types include:

Lever-Actuated Switches:

- Suitable for gradual approach detection
- Operating force: 50-300 grams
- Pretravel: 0.5-2.0 mm
- Repeatability: ± 0.01 mm
- Contact rating: 10-20 A at 125/250 VAC

Roller-Plunger Switches:

- Provide consistent actuation force
- Reduced mechanical wear compared to lever types
- Ideal for high-cycle applications

Magnetic Reed Switches:

- Non-contact operation for harsh environments
- Sealed contacts protect against dust and moisture
- Operating distance: 5-20 mm from magnet

Application in PCB Production Line:

- X, Y, Z axis homing for CNC station
- Conveyor end-of-travel detection
- Flipping mechanism position verification
- Emergency stop interlock circuits

2.6.2 Proximity Sensors

Inductive proximity sensors detect metallic objects without physical contact, useful for PCB presence detection and position verification.

Sensor Characteristics:

- **Sensing Range:** 2-15 mm for standard inductive sensors
- **Output Type:** NPN or PNP transistor, NO or NC configuration
- **Supply Voltage:** 6-36 V DC
- **Switching Frequency:** 100-2000 Hz
- **Target Material:** Ferrous metals (reduced range for non-ferrous)

Capacitive Proximity Sensors:

- Detect both metallic and non-metallic materials
- Suitable for PCB substrate detection (FR-4, phenolic)
- Sensing range: 2-10 mm
- Less affected by target material composition

Application Examples:

```
1 // PCB detection using inductive sensor
2 const int pcbSensor = 3;
3 bool pcbPresent = false;
4
5 void checkPCBPresence() {
6     pcbPresent = (digitalRead(pcbSensor) == LOW); // Active LOW
7     if (pcbPresent) {
8         startProcessingSequence();
9     }
10 }
11
12 // Conveyor jam detection using dual sensors
13 const int sensor1 = 4, sensor2 = 5;
14 unsigned long lastTrigger = 0;
15 const unsigned long maxInterval = 5000; // 5 seconds
16
17 void monitorConveyor() {
18     if (digitalRead(sensor1) == LOW) {
19         lastTrigger = millis();
20     }
21     if (millis() - lastTrigger > maxInterval) {
22         triggerJamAlarm();
23     }
24 }
```

Figure 2.15: Proximity Sensor

2.6.3 Optical Sensors

Photoelectric sensors use light beams for object detection, offering longer sensing ranges and versatility. Types include:

Through-Beam Sensors:

- Separate emitter and receiver, longest range (up to 10 m)
- Highest reliability and detection accuracy
- Ideal for critical safety interlocks

Retroreflective Sensors:

- Emitter and receiver in one housing with reflector
- Medium range (0.1-5 m)
- Easier installation than through-beam type

Diffuse Reflective Sensors:

- Detect light reflected from target object
- Short range (2-100 mm)
- Suitable for PCB edge detection and alignment

Applications in PCB Production:

- Board presence detection on conveyor
- Edge alignment verification before drilling
- Component counting in assembly station
- Spray nozzle clog detection (via light scattering)

2.6.4 Fluid Level and Pressure Sensors

For the automated spraying system, fluid monitoring sensors are essential :

Liquid Level Sensors:

- **Float Switches:** Mechanical, reliable, low cost
- **Capacitive Sensors:** Non-contact, suitable for corrosive fluids
- **Ultrasonic Sensors:** Continuous level measurement, high accuracy

Pressure Sensors:

- Monitor spray system pressure for consistent atomization
- Typical range: 0-10 bar for pneumatic spray systems
- Output: 4-20 mA or 0-5 V analog signal
- Accuracy: $\pm 0.5\%$ full scale

Integration with Arduino:

```
1 // Analog pressure sensor reading
2 const int pressurePin = A0;
3 float pressure = 0;
4 const float minPressure = 2.0; // bar
5 const float maxPressure = 6.0; // bar
6
7 void readPressure() {
8     int sensorValue = analogRead(pressurePin);
9     pressure = map(sensorValue, 0, 1023, 0, 10); // 0-10 bar range
10
11     if (pressure < minPressure || pressure > maxPressure) {
12         triggerPressureAlarm();
13     }
14 }
```

Figure 2.16: Integration with Arduino

2.7 POWER SUPPLY AND DISTRIBUTION

2.7.1 Voltage Regulation Requirements

PCB production line systems require multiple voltage levels for different subsystems:

Table 2-1: Voltage Regulation Requirements

Voltage Level	Application	Current Requirement
5V DC	Arduino logic, digital sensors, servo control	500 mA - 2 A
12V DC	Stepper drivers, relay coils, indicator LEDs	2-10 A
24V DC	Industrial sensors, solenoid valves, high-torque motors	5-20 A
120/240V AC	CNC spindle motor, main power distribution	10-30 A

2.7.2 Switching Power Supplies

Switching power supplies offer efficient voltage conversion with compact size :

Key Specifications:

- **Efficiency:** 80-90% typical
- **Regulation:** ± 1 -2% line and load regulation
- **Ripple:** < 50 mV peak-to-peak
- **Protection:** Over-current, over-voltage, short-circuit protection
- **Power Factor Correction (PFC):** Required for supplies > 75 W (IEC 61000-3-2)

Selection Criteria for PCB Production Line:

Total Power Calculation:

- Arduino + Sensors: $5V \times 1A = 5W$
- Stepper Drivers ($\times 3$): $12V \times 3A \times 3 = 108W$
- Servo Motors ($\times 2$): $6V \times 2A \times 2 = 24W$
- Spray System Pump: $24V \times 2A = 48W$
- CNC Spindle: $120V \times 5A = 600W$
- Safety Margin (20%): $\sim 157W$
- Total Recommended: $\sim 950W$

Recommended Configuration:

- 5V/3A switching supply (for logic)
- 12V/10A switching supply (for motors)
- 24V/3A switching supply (for spray system)
- Separate AC circuit for CNC spindle

2.7.3 Power Distribution and Safety

Distribution Architecture:

- Main AC input → Circuit breaker → EMI filter → Distribution bus
- DC supplies connected to common ground bus
- Star topology for ground connections to minimize ground loops

Safety Features :

- **Emergency Stop Circuit:** Hardwired NC contacts, Category 0 stop per IEC 60204-1
- **Fusing:** Individual fuses for each subsystem (fast-blow for electronics, slow-blow for motors)
- **Isolation:** Opto-isolators for signal lines between high/low voltage sections
- **Grounding:** Single-point ground for control circuits, chassis ground for safety

Wiring Guidelines:

- Signal cables: 22-24 AWG, shielded for analog signals
- Power cables: 16-18 AWG for 12/24V DC, 14 AWG for AC mains
- Cable routing: Separate power and signal cables to reduce EMI
- Connectors: Screw terminals for permanent connections, Molex/JST for modular sections

2.8 PREVIOUS WORK AND RELATED RESEARCH

2.8.1 Academic Research in PCB Automation

Several research projects have explored automated PCB manufacturing:

Kim and Park (2018) developed a modular CNC drilling system for PCB prototyping using FPGA-based motion control, achieving positioning accuracy of ± 0.015 mm and drilling speeds up to 150 holes per minute.

Hassan and Ali (2020) implemented an Arduino-based automation system for small-scale electronics manufacturing, demonstrating cost reduction of 40% compared to manual methods while maintaining acceptable quality standards.

Gupta and Chen (2021) designed a closed-loop servo control system for precision positioning in automated assembly, utilizing PID control with feedforward compensation to achieve tracking errors below 0.05 mm.

Recent Work on Coating Systems:

- **Lee et al. (2022)** investigated spray coating parameters for conformal coating on PCBs, optimizing nozzle pressure and standoff distance for uniform thickness.
- **Martinez and Wong (2023)** developed a low-cost automated dispensing system for small-batch PCB production using stepper-driven syringe pumps.

2.8.2 Industrial PCB Manufacturing Systems

Commercial PCB production lines from companies like LPKF, Baccini, and Ekra offer fully automated solutions with features including:

- Automated optical inspection (AOI) with machine vision
- Laser drilling for microvias (< 0.15 mm diameter)
- Robotic material handling and board transfer
- Integration with Manufacturing Execution Systems (MES)
- Real-time process monitoring and statistical process control (SPC)

However, these systems are cost-prohibitive for educational institutions and small-scale manufacturers, with prices ranging from \$50,000 to over \$500,000.

2.8.3 Open-Source CNC and Automation Projects

The open-source hardware movement has produced several relevant projects:

GRBL: Open-source CNC control firmware for Arduino, supporting G-code interpretation and stepper motor control.

Universal G-code Sender (UGS): Cross-platform Java application for CNC machine control.

Easel: Web-based CNC design and control platform by Inventables.

OpenBuilds: Modular linear motion systems for DIY CNC machines.

These projects demonstrate the feasibility of low-cost, Arduino-based automation but lack integration into complete production line systems with spraying, flipping, and assembly stages.

2.9 STANDARDS AND SAFETY REQUIREMENTS

2.9.1 Electrical Safety Standards

IEC 60204-1 "Safety of Machinery – Electrical Equipment of Machines" specifies requirements for electrical safety in industrial equipment:

- Emergency stop circuits must be Category 0 or Category 1 stops
- Protection against electric shock (insulation, grounding, enclosures)
- Overcurrent and short-circuit protection
- Control circuit safety and reliability
- Cable and conductor sizing

Key Requirements for This Project:

- Emergency Stop: Hardwired NC circuit, manual reset required
- Enclosure Rating: IP54 minimum for spray chamber (dust/water protection)
- Grounding: All metal parts bonded to protective earth
- Isolation: 2.5 kV dielectric strength between primary and secondary circuits
- Labelling: Clear voltage and hazard warnings on all accessible panels

2.9.2 PCB Manufacturing Standards

IPC (Association Connecting Electronics Industries) standards govern PCB fabrication:

Table 2-2:IPC Standard

Standard	Scope	Relevant Parameters
IPC-A-600	Acceptability of Printed Boards	Visual inspection criteria, defect classification
IPC-6012	Qualification for Rigid PCBs	Mechanical strength, thermal stress resistance
IPC-2221	Generic Standard on PCB Design	Minimum trace width, clearance, annular ring
IPC-J-STD-001	Requirements for Soldered Assemblies	Solder joint quality, component placement

Key Parameters from These Standards:

- Minimum annular ring: 0.15 mm (Class 2), 0.25 mm (Class 3)
- Minimum drill-to-copper clearance: 0.20 mm
- Solder mask registration: ± 0.05 mm
- Surface insulation resistance: $> 100 \text{ M}\Omega$
- Coating thickness uniformity: $\pm 10\%$ of nominal value

2.9.3 Environmental and Material Safety

Spray Coating Safety Considerations:

- Ventilation: Minimum 10 air changes/hour in spray chamber
- PPE: Respirator, gloves, eye protection for operators
- Material SDS: Safety Data Sheets for all coating materials
- Waste disposal: Compliant with local hazardous waste regulations

Flammability and Fire Safety:

- Coating materials: Flash point $> 60^\circ\text{C}$ preferred
- Electrical components: UL 94 V-0 flame rating for enclosures
- Fire suppression: CO_2 or dry chemical extinguisher nearby

2.10 SUMMARY

This chapter has presented the theoretical foundations necessary for understanding and designing an automated PCB production line. Key topics covered include:

1. **PCB Manufacturing Fundamentals:** Structure, fabrication steps, and insulation coating techniques essential for electrical isolation and process reliability.
2. **Automated Spraying Systems:** Principles of atomization, nozzle selection criteria, fluid dynamics considerations, and drying/curing processes critical for uniform insulation layer application.
3. **CNC Technology:** System architecture, G-code programming, and spindle specifications enabling precision drilling and routing operations.
4. **Embedded Control Systems:** Arduino platform capabilities, real-time control considerations, and firmware development strategies for coordinating multi-stage production processes.
5. **Sensor Technologies:** Limit switches, proximity sensors, optical sensors, and fluid monitoring systems for process feedback and safety interlocks.
6. **Power Supply and Distribution:** Voltage requirements, switching power supply selection, and safety-oriented distribution architecture.
7. **Previous Work:** Review of academic research, industrial systems, and open-source projects highlighting the gap this project addresses.
8. **Standards and Safety:** Electrical safety (IEC 60204-1), PCB manufacturing standards (IPC), and environmental considerations for spray coating operations.

The literature review reveals that while industrial-grade PCB manufacturing systems offer high performance, they are cost-prohibitive for small-scale applications. Conversely, open-source projects demonstrate the viability of Arduino-based control but lack integration into complete production workflows with spraying, flipping, and assembly stages. This project addresses this gap by developing an integrated, cost-effective PCB production line that combines embedded control, precision mechanics, automated spraying, and process sequencing suitable for educational and small-scale manufacturing environments.

The next chapter will detail the specific design methodology and system architecture developed based on these theoretical foundations.

CHAPTER 3: System Design and Methodology



Figure 3.1: System Design and Methodology

3.1 SYSTEM OVERVIEW

The PCB production line is an integrated mechatronic system designed to automate the manufacturing process of printed circuit boards through five sequential stages: insulation coating application, CNC drilling and routing, PCB flipping, component marking, and assembly preparation. The system employs a modular design approach utilizing aluminum profile framing (3030 and 2020 series) to provide structural rigidity while maintaining ease of assembly and modification.

The overall system architecture consists of two main workstations, each with a 1-meter length frame, connected by a timing belt conveyor system (6mm width). Each workstation houses specific processing mechanisms controlled by dedicated NEMA 17 stepper motors for precision motion control and DC geared motors for conveyor transport. The control system is based on Arduino microcontrollers coordinating multiple subsystems including servo-driven flipping mechanisms (SG90 and MG995), CNC motion systems with both lead screw and belt drive configurations, and an automated spray coating station.

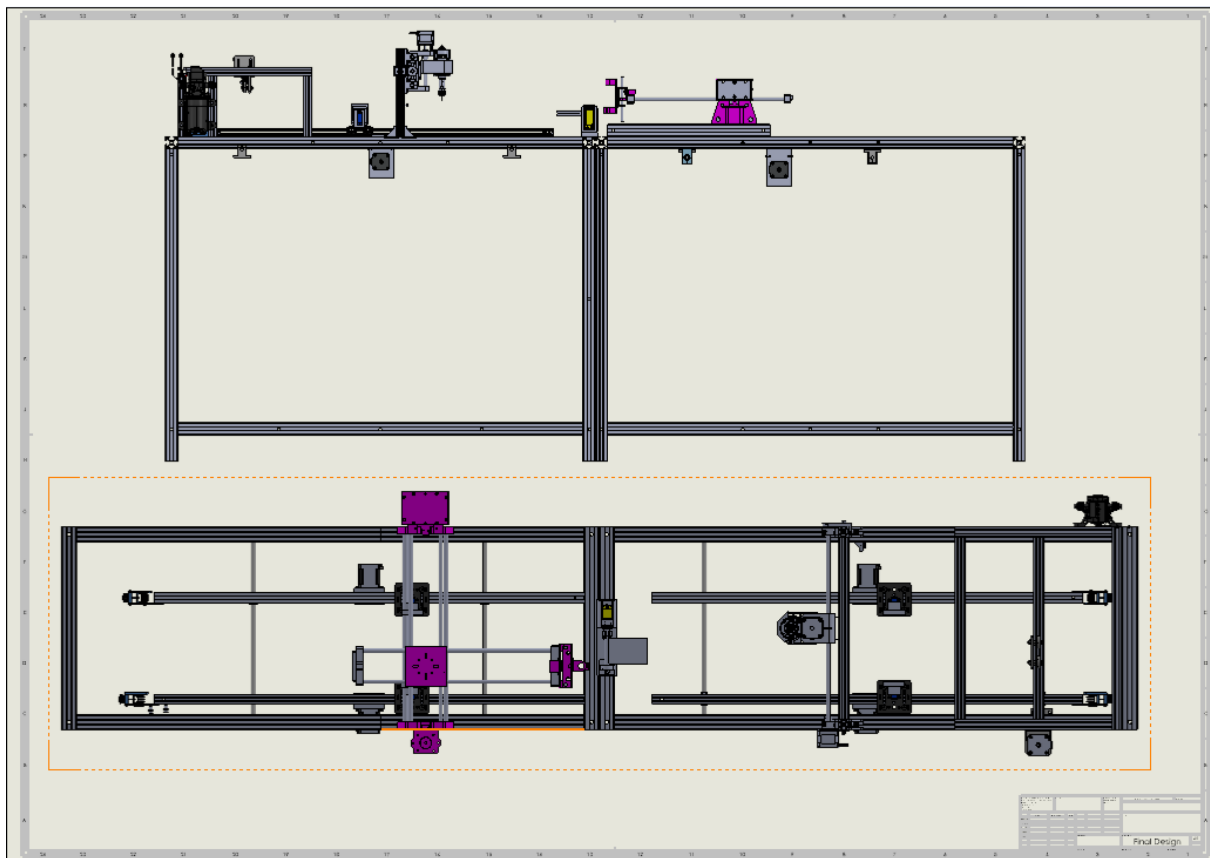


Figure 3.2: System Overview

The system operates as follows: Raw PCB substrates are loaded onto the conveyor belt and transported to the spraying chamber where a compressor-driven spray gun applies insulating coating in a two-axis motion pattern. The board then proceeds to the first CNC station equipped with a lead screw-driven mechanism for precision drilling and routing operations. Upon completion, a U-shaped 3D-printed flipping mechanism engages the PCB, closes securely using SG90 servos, and rotates it 180° to enable double-sided processing. The board then moves to the second CNC station with belt-driven axes for component marking using a pen plotter attachment. Finally, the processed PCB reaches the assembly station for component placement.

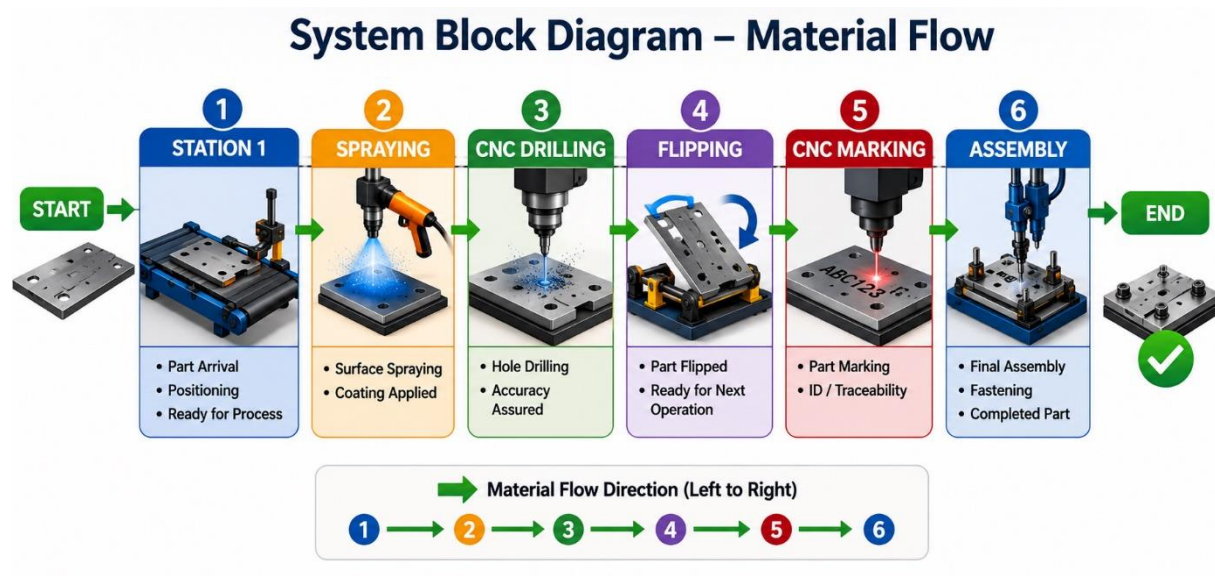


Figure 3.3: System Block Diagram

3.2 DESIGN REQUIREMENTS

3.2.1 Functional Requirements:

1. Conveyor Transport System:

- Transport PCBs between stations at controlled speed (50-150 mm/s)
- Maintain consistent spacing between multiple boards
- Support PCB sizes up to 200mm × 150mm
- Provide lateral alignment using guide rails

2. Spraying Chamber:

- Apply uniform insulation coating (25-50 μm thickness)
- Two-axis motion control (X-Y plane, no Z-axis)
- Belt-driven positioning system
- Compressor-operated spray gun with adjustable pressure (2-4 bar)
- Drying time: 5-10 minutes

3. CNC Drilling/Routing Station:

- Positioning accuracy: ± 0.02 mm
- Lead screw-driven axes for high precision
- Spindle speed: 12,000-24,000 RPM
- Drill bit range: 0.3-3.0 mm diameter
- Maximum travel: 300mm (X) × 200mm (Y) × 50mm (Z)

4. PCB Flipping Mechanism:

- 180° rotation capability
- Secure gripping during conveyor motion
- Cycle time: < 3 seconds
- Compatible with PCB thickness: 0.8-1.6 mm
- Dual servo actuation (SG90 for gripping, MG995 for rotation)

5. CNC Marking Station:

- Belt-driven axes for faster operation
- Pen plotter attachment for silkscreen marking
- Positioning accuracy: ± 0.05 mm
- Marking speed: 100-200 mm/s

6. Control System:

- Arduino UNO as main controller
- Real-time coordination of all subsystems
- Emergency stop functionality
- Limit switch homing for all axes

3.2.2 Non-Functional Requirements:

1. Structural Requirements:

- Frame rigidity: Maximum deflection < 0.1 mm under load
- Vibration damping for precision operations
- Modular design for easy maintenance
- Overall dimensions: 2000mm (L) $\times$ 600mm (W) $\times$ 800mm (H)

2. Performance Requirements:

- Production throughput: 10-15 boards per hour
- System availability: $> 95\%$
- Mean time between failures (MTBF): > 500 hours
- Power consumption: < 500 W total

3. Safety Requirements:

- Emergency stop circuits (Category 0 stop per IEC 60204-1)
- Overcurrent protection for all motors
- Thermal protection for spindle motor
- Enclosed moving parts to prevent operator injury
- Proper grounding and electrical isolation

4. Material Requirements:

- Aluminium profiles: 6063-T5 alloy (3030 and 2020 series)
- 3D printed parts: PLA or PETG filament
- Standard components: LM8UU linear bearings, M3-M8 fasteners
- Timing belt: GT2 profile, 6mm width

3.2.3 System Specifications Summary

Table 3-1: System Specification Summary

Parameter	Specification	Unit
Overall System Dimensions		
Total Length	2000	mm
Total Width	4000	mm
Total Height	8000	mm
Frame Structure		
Main Profile	3030 (30×30)	mm
Secondary Profile	2020 (20×20)	mm
Material	Aluminium 6063-T5	-
Conveyor System		
Belt Type	GT2 Timing Belt	-
Belt Width	6	mm
Transport Speed	50-150	mm/s
Motor Type	DC Geared Motor	-
CNC Drilling Station		
Drive System	Lead Screw (T8)	-
Lead	2	mm/rev
X-Axis Travel	300	mm
Y-Axis Travel	200	mm
Z-Axis Travel	50	mm

Positioning Accuracy	± 0.02	mm
Spindle Speed	12,000-24,000	RPM
Motor Type	NEMA 17 Stepper	-
Spray Coating Station		
Drive System	GT2 Timing Belt	--
X-Axis Travel	400	mm
Y-Axis Travel	300	mm
Nozzle Diameter	0.3	mm
Operating Pressure	2-4	bar
Coating Thickness	25-50	μm
PCB Flipping Mechanism		
Rotation Angle	180	degrees
Gripping Servo	SG90	-
Rotation Servo	MG995	-
Cycle Time	< 3	seconds
CNC Marking Station		
Drive System	GT2 Timing Belt	--
X-Axis Travel	300	mm
Y-Axis Travel	200	mm
Marking Speed	100-200	mm/s
Positioning Accuracy	± 0.05	mm
Control System		

Main Controller	Arduino UNO	-
Operating Voltage	12/5	V DC
Total Power Consumption	< 500	W
Performance		
Production Throughput	10-15	boards/hour
Supported PCB Size (max)	200×150	mm
PCB Thickness Range	0.8-1.6	mm

3.3 MECHANICAL DESIGN

3.3.1 Frame Structure Design

The production line frame is constructed using modular Aluminium extrusion profiles (6063-T5 alloy) in two standard sizes: 3030 (30mm × 30mm) and 2020 (20mm × 20mm). The selection of profile size is based on structural load requirements:

3030 Profile Applications:

- Main vertical supports and load-bearing members
- Base frame structure for both workstations
- CNC machine mounting points
- Maximum span: 1000mm between supports

2020 Profile Applications:

- Secondary supports and cross-bracing
- Sensor mounting brackets
- Cable management structures
- Guide rail supports

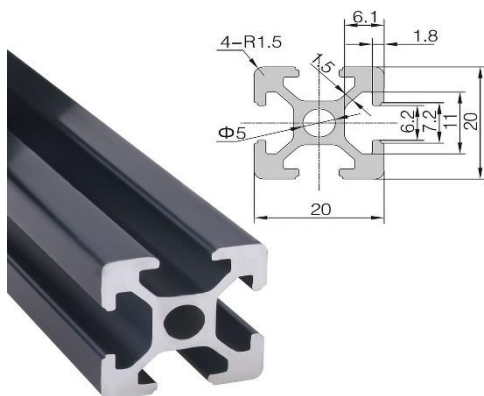


Figure 3.5:2020 Profile Aluminum

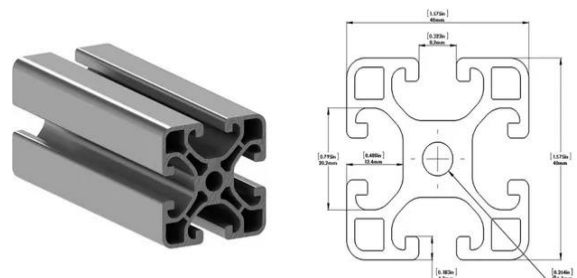


Figure 3.4:3030 Profile Aluminum

Design Calculations:

Maximum Deflection Calculation for 3030 Profile:

For a simply supported beam with central point load: $\delta_{max} = \frac{FL^3}{48EI}$

Where:

- F = Maximum load = 50 N (including motor, spindle, and dynamic forces)
- L = Span length = 1000 mm = 1.0 m
- E = Modulus of Elasticity (6063-T5) = 68.9 GPa = 68.9×10^9 Pa
- I = Moment of inertia for 3030 profile $\approx 2.5 \times 10^{-8} \text{ m}^4$

$$\delta_{max} = \frac{50 \times (1.0)^3}{48 \times 68.9 \times 10^9 \times 2.5 \times 10^{-8}} \delta_{max} = \frac{48 \times 68.9 \times 10^9 \times 2.5 \times 10^{-8}}{850 \times (1.0)^3}$$

$$\delta_{max} = 0.061 \text{ mm} < 0.1 \text{ mm (requirement satisfied)}$$

$$\delta_{max} = 0.061 \text{ mm} < 0.1 \text{ mm (requirement satisfied)}$$

Frame Weight Estimation:

- 3030 profile weight: 0.82 kg/m
- 2020 profile weight: 0.37 kg/m
- Total 3030 length: $\approx 12 \text{ m} \rightarrow 9.84 \text{ kg}$
- Total 2020 length: $\approx 8 \text{ m} \rightarrow 2.96 \text{ kg}$
- **Total frame weight: $\approx 13 \text{ kg}$**

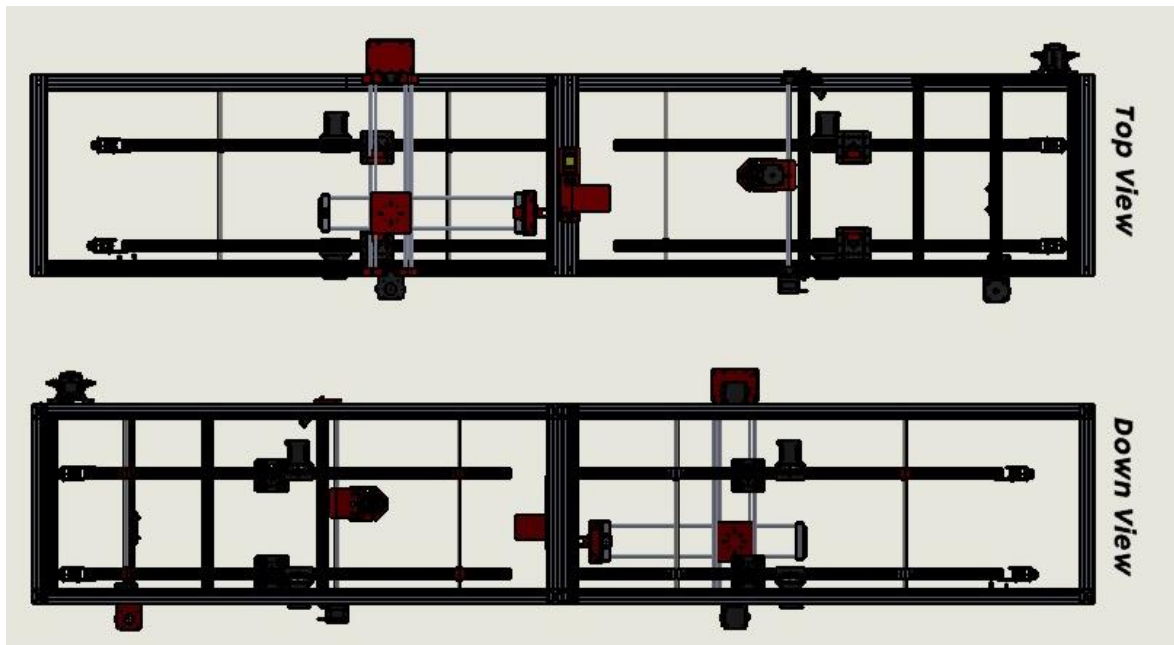


Figure 3.6: Top and Down View

3.3.2 Conveyor System Design

The conveyor system employs a GT2 timing belt (6mm width) driven by a DC geared motor to transport PCBs between processing stations. The timing belt profile ensures positive engagement and prevents slippage during transport.

Component Selection:

DC Geared Motor Specifications:

- Operating voltage: 12V DC
- No-load speed: 100 RPM (output shaft)
- Stall torque: 2.0 kg·cm
- Gear ratio: 1:48
- Power rating: 6W

Belt Speed Calculation:

$$v = \pi \times D \times N / 60 \quad v = 60 \pi \times D \times N$$

Where:

- D = Pulley diameter = 20 mm = 0.02 m
- N = Motor speed = 100 RPM

$$v = \pi \times 0.02 \times 100 / 60 = 0.105 \text{ m/s} = 105 \text{ mm/s} \quad v = 60 \pi \times 0.02 \times 100 = 0.105 \text{ m/s} = 105 \text{ mm/s}$$

This speed falls within the required range of 50-150 mm/s.

Motor Power Requirement:

$$P = F \times v \quad P = F \times v$$

Assuming:

- F = Total friction force = 2 N (belt friction + PCB weight)
- $v = 0.105 \text{ m/s}$

$$P = 2 \times 0.105 = 0.21 \text{ W} \quad P = 2 \times 0.105 = 0.21 \text{ W}$$

The selected 6W motor provides adequate power with safety factor of ≈ 28 .

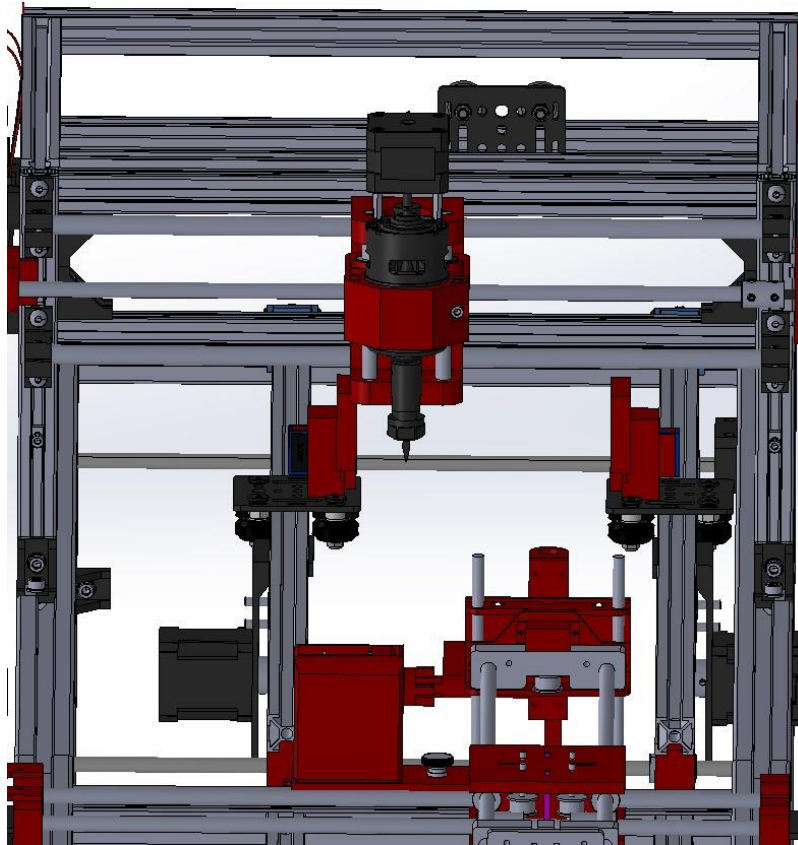


Figure 3.9: Conveyor System with Belt Shown

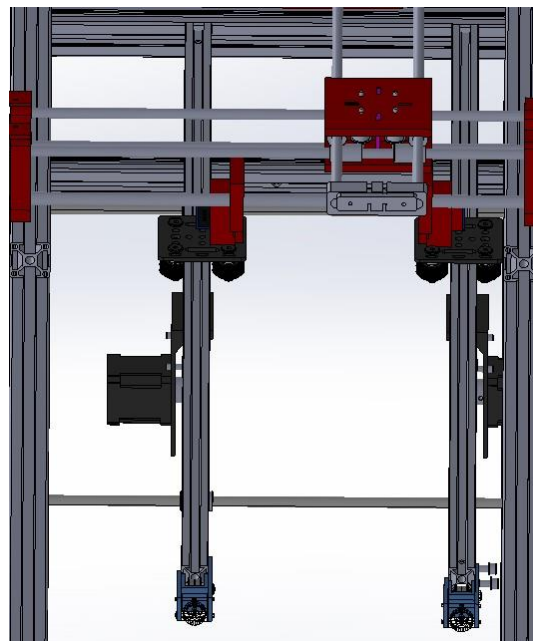


Figure 3.8Belt System with Mechanism

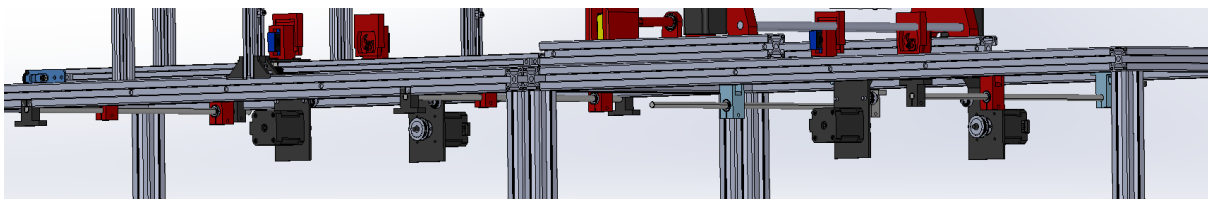


Figure 3.7:Motor Mounting Path

3.3.3 PCB Flipping Mechanism Design

The flipping mechanism is a critical subsystem that enables double-sided PCB processing through automated 180° rotation. The design utilizes 3D-printed components (PLA/PETG) actuated by servo motors.

Mechanical Design:

U-Shaped Gripper:

- 3D-printed U-profile that engages PCB edges
- Inner width: 210mm (accommodates PCBs up to 200mm)
- Gripping force: Provided by two SG90 servos
- Material: PETG filament (better impact resistance than PLA)

Servo Motor Selection:

SG90 Micro Servo (Gripping):

- Operating voltage: 4.8-6.0V
- Stall torque: 1.8 kg·cm at 4.8V
- Speed: 0.12 sec/60°
- Weight: 9g
- Quantity: 2 units (dual-sided gripping)

MG995 High-Torque Servo (Rotation):

- Operating voltage: 4.8-7.2V
- Stall torque: 10 kg·cm at 6.0V
- Speed: 0.17 sec/60°
- Weight: 55g
- Quantity: 1 unit

Torque Calculation for Rotation:

$$T_{required} = m \times g \times r$$

Where:

- m = Mass of PCB + gripper = 0.15 kg (PCB: 0.05 kg, gripper: 0.10 kg)
- $g = 9.81 \text{ m/s}^2$
- r = Distance from rotation axis to center of mass = 0.05 m

$$T_{required} = 0.15 \times 9.81 \times 0.05 = 0.074 \text{ N}\cdot\text{m} = 0.75 \text{ kg}\cdot\text{cm}$$

$$= 0.15 \times 9.81 \times 0.05 = 0.074 \text{ N}\cdot\text{m} = 0.75 \text{ kg}\cdot\text{cm}$$

MG995 provides 10 kg·cm torque, giving a safety factor of **13.3** (adequate for dynamic loading).

Rotation Time: For 180° rotation: $t = 180^\circ / 60^\circ \times 0.17 \text{ sec} = 0.51 \text{ sec}$

Adding acceleration/deceleration: **Total cycle time $\approx 1.5 \text{ sec}$** (well within 3 sec requirement)

Locking Mechanism: The mechanism includes 3D-printed latches that engage the PCB edges while on the conveyor, preventing displacement during transport. SG90 servos control the latch opening/closing synchronized with conveyor motion.

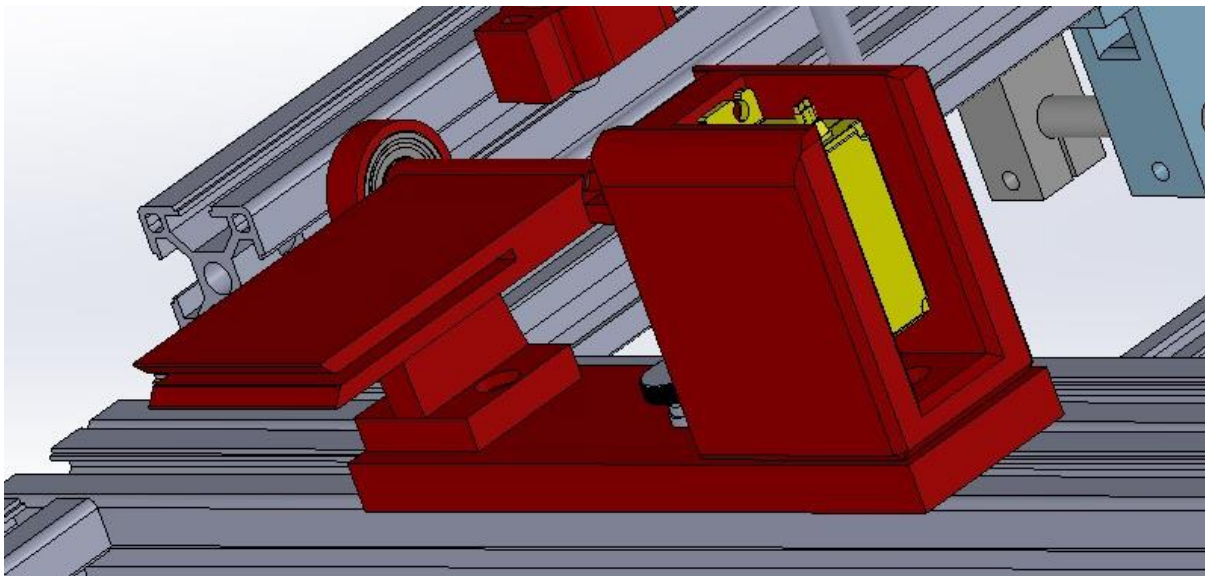


Figure 3.10: Flipping Mechanism

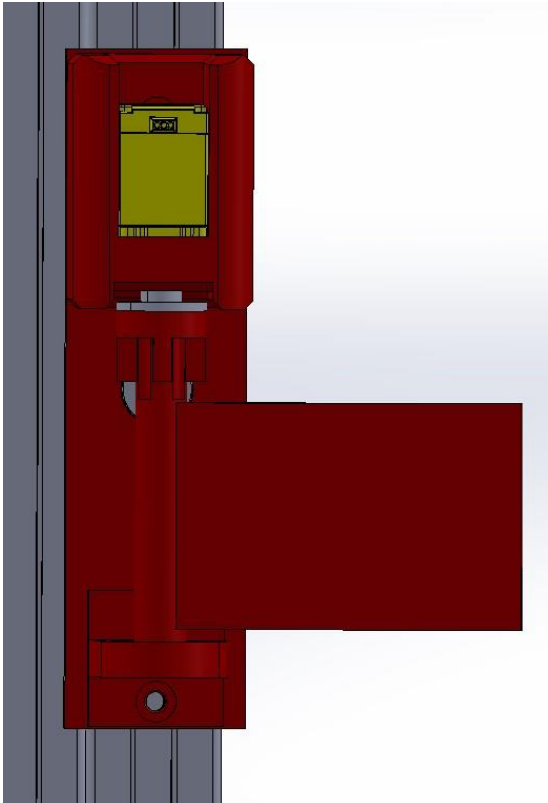


Figure 3.14: Flipping Mechanism Top View

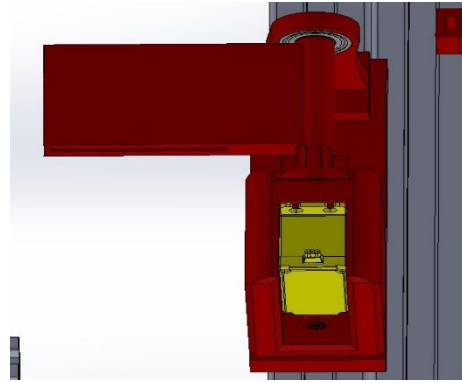


Figure 3.13: Flipping Mechanism Isometric View

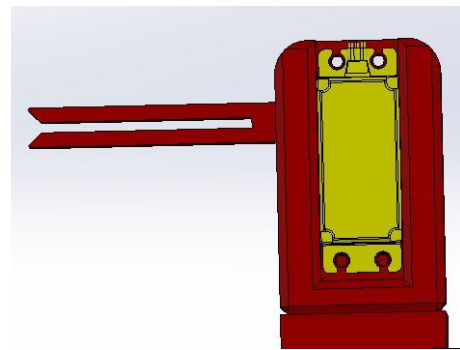


Figure 3.12: Flipping Mechanism Side View

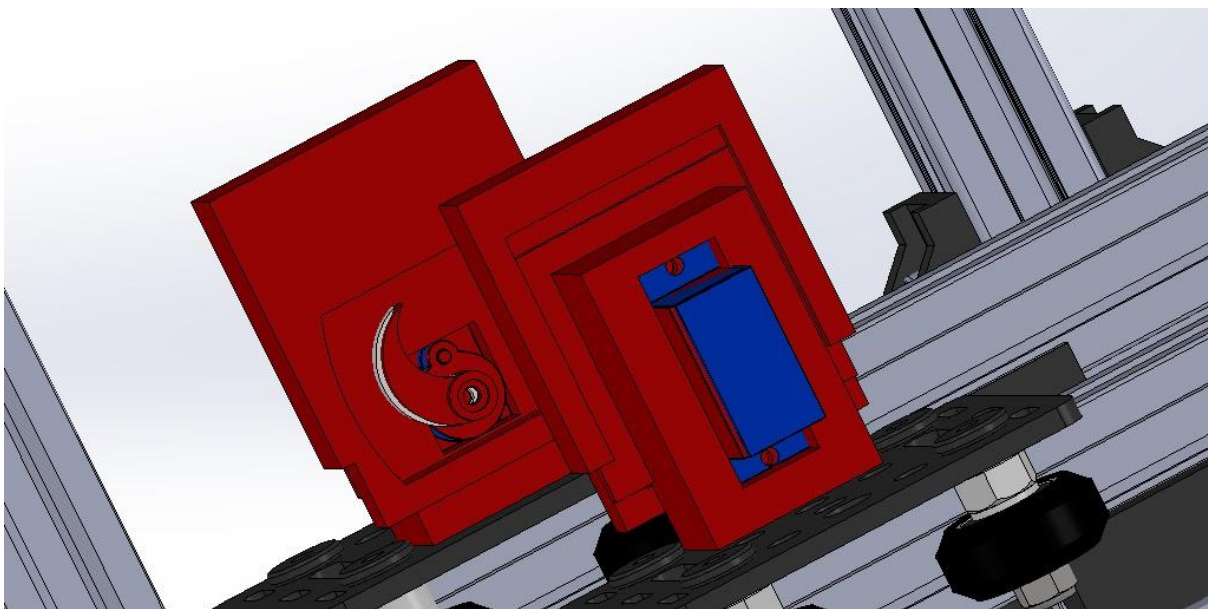


Figure 3.11: Lock Mechanism

3.3.4 CNC Drilling and Routing Station Design

The first CNC station performs precision drilling and routing operations using a lead screw-driven motion system for high accuracy.

Motion System Design:

Lead Screw Drive:

- Lead screw specification: T8 (8mm diameter, 2mm pitch)
- Lead: 2mm/revolution
- Nut type: Brass anti-backlash nut
- Efficiency: $\eta \approx 0.40$ (40%)

Stepper Motor Selection (NEMA 17):

- Holding torque: 0.5 N·m (5.1 kg·cm)
- Step angle: 1.8° (200 steps/revolution)
- Rated current: 1.7 A
- Operating voltage: 12V
- Microstepping: 1/16 (3200 steps/revolution)

Positional Resolution:

$\text{Resolution} = \frac{\text{Lead}}{\text{Steps/Rev}} = \frac{2 \text{ mm}}{3200} = 0.000625 \text{ mm} = 0.625 \text{ }\mu\text{m}$
 $\text{Resolution} = \frac{\text{Steps/Rev}}{\text{Lead}} = \frac{3200}{2 \text{ mm}} = 0.000625 \text{ mm} = 0.625 \text{ }\mu\text{m}$

Travel Range:

- X-axis: 300 mm
- Y-axis: 200 mm
- Z-axis: 50 mm

Maximum Linear Speed:

Assuming maximum motor speed = 300 RPM:

$$v_{max} = \text{Lead} \times N = 2 \text{ mm/rev} \times 300 \text{ rev/min} = 600 \text{ mm/min} = 10 \text{ mm/s}$$
$$= \text{Lead} \times N = 2 \text{ mm/rev} \times 300 \text{ rev/min} = 600 \text{ mm/min} = 10 \text{ mm/s}$$

Torque Requirement for X-Axis:

$$T = F \times \text{Lead} / 2\pi \times \eta \quad T = 2\pi \times \eta F \times \text{Lead}$$

Where:

- F = Cutting force + friction = 15 N (estimated for PCB drilling)
- $\text{Lead} = 2 \text{ mm} = 0.002 \text{ m}$
- $\eta = 0.40$

$$T = 15 \times 0.002 / 2\pi \times 0.40 = 0.012 \text{ N}\cdot\text{m} = 0.12 \text{ kg}\cdot\text{cm}$$
$$T = 2\pi \times 0.40 \times 15 \times 0.002 = 0.012 \text{ N}\cdot\text{m} = 0.12 \text{ kg}\cdot\text{cm}$$

NEMA 17 provides 5.1 kg·cm, giving safety factor of **42.5** (more than adequate).

Spindle Motor:

- Type: DC brushless spindle
- Power: 775 DC motor
- Speed range: 12,000-24,000 RPM (PWM controlled)
- Collet: ER11 (supports 0.5-7mm drill bits)
- Mounting: Aluminum bracket with vibration damping

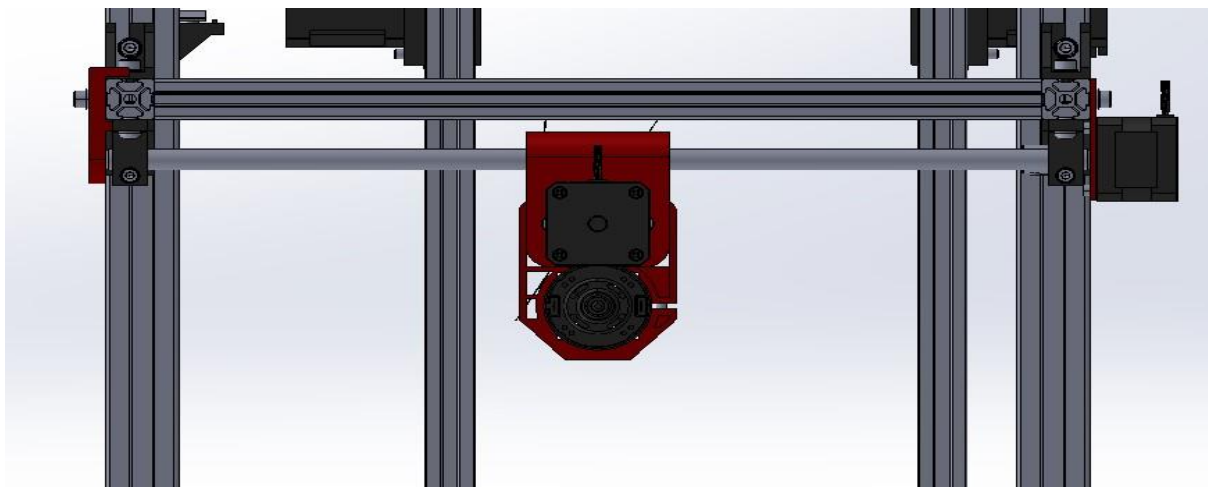


Figure 3.15: CNC Drilling Station Top View

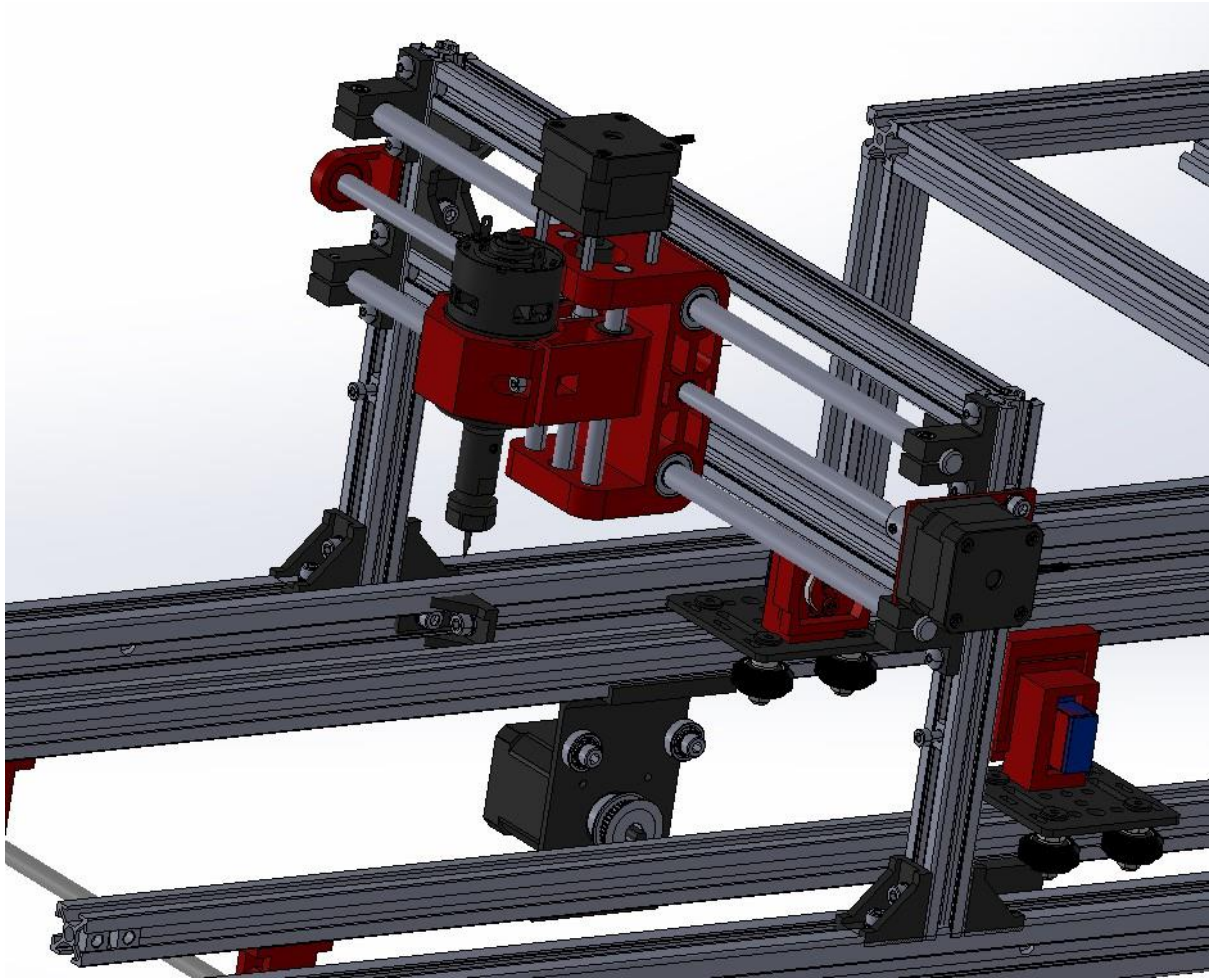


Figure 3.17:Cnc Drilling Station Isometric

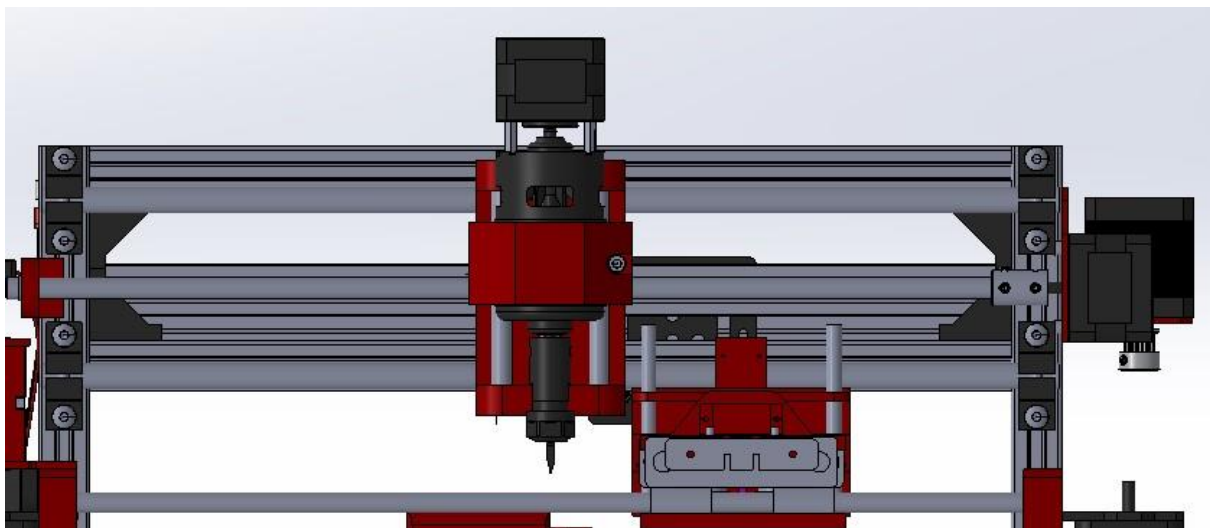


Figure 3.16:Cnc Drilling Front View

3.3.5 Spray Coating Chamber Design

The spraying chamber applies insulation coating using a compressor-operated spray gun mounted on a two-axis (X-Y) belt-driven positioning system.

Mechanical Design:

Belt-Driven Motion System:

- Belt type: GT2 timing belt (6mm width)
- Pulley diameter: 16 teeth (10.2mm pitch diameter)
- Drive configuration: Dual NEMA 17 steppers (one per axis)
- No Z-axis (fixed standoff distance)

Travel Range:

- X-axis: 400 mm
- Y-axis: 300 mm
- Z-standoff: 150 mm (fixed)

Linear Resolution:

For GT2 belt with 16-tooth pulley:

Linear travel per revolution = $\pi \times D = \pi \times 10.2 = 32$ mm
Linear travel per revolution = $\pi \times D = \pi \times 10.2 = 32$ mm

With 1/16 microstepping (3200 steps/rev):

Resolution = $32 \text{ mm} / 3200 = 0.01 \text{ mm} = 10 \text{ }\mu\text{m}$
Resolution = $32 / 3200 = 0.01 \text{ mm} = 10 \text{ }\mu\text{m}$

Spray Gun Mounting:

- Compressor-operated spray gun (airbrush type)
- Operating pressure: 2-4 bar
- Nozzle diameter: 0.3mm
- Flow rate: 5-15 mL/min
- Mounting: 3D-printed bracket with adjustable angle

Spray Pattern Control:

The spray gun follows a raster pattern:

- Line spacing: 5mm (overlap for uniform coverage)
- Traverse speed: 50 mm/s
- Number of passes: 2-3 for desired thickness

Coating Thickness Estimation:

Thickness=Flow rate×Number of passesTraverse speed×Spray width
 $Thickness = \frac{Flow\ rate \times Number\ of\ passes}{Traverse\ speed \times Spray\ width}$

Assuming:

- Flow rate = 10 mL/min = 167 mm³/s
- Traverse speed = 50 mm/s
- Spray width = 20 mm
- Passes = 2

Thickness= $\frac{167 \times 250 \times 20}{50 \times 20} = 0.334\text{ mm} = 334\text{ }\mu\text{m}$

This exceeds the requirement (25-50 μm), so flow rate or passes can be reduced.

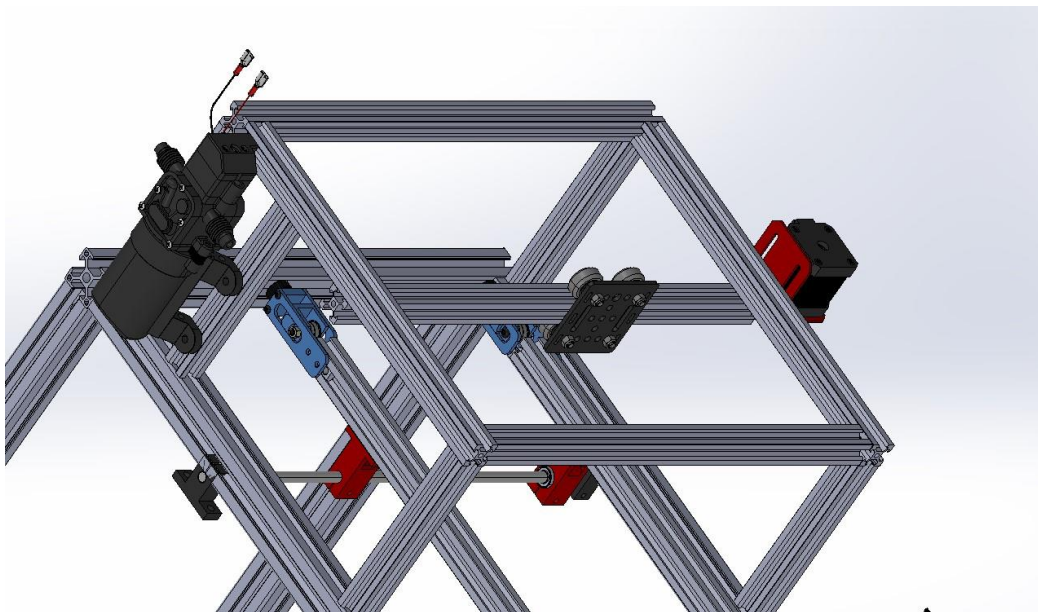


Figure 3.18:Painting Station

3.3.6 CNC Marking Station (Pen Plotter) Design

The second CNC station performs component marking using a pen plotter attachment with belt-driven axes for faster operation compared to the drilling station.

Motion System:

Belt Drive Configuration:

- Same GT2 belt system as spray chamber
- Dual NEMA 17 steppers (X-Y axes)
- Travel range: 300mm (X) × 200mm (Y)
- Resolution: 10 μ m (same calculation as spray chamber)

Pen Plotter Mechanism:

- Servo-actuated pen lift (SG90)
- Pen holder: 3D-printed clamp (accommodates 8-12mm diameter pens)
- Drawing force: 0.5-1.0 N (controlled by servo position)
- Maximum marking speed: 200 mm/s

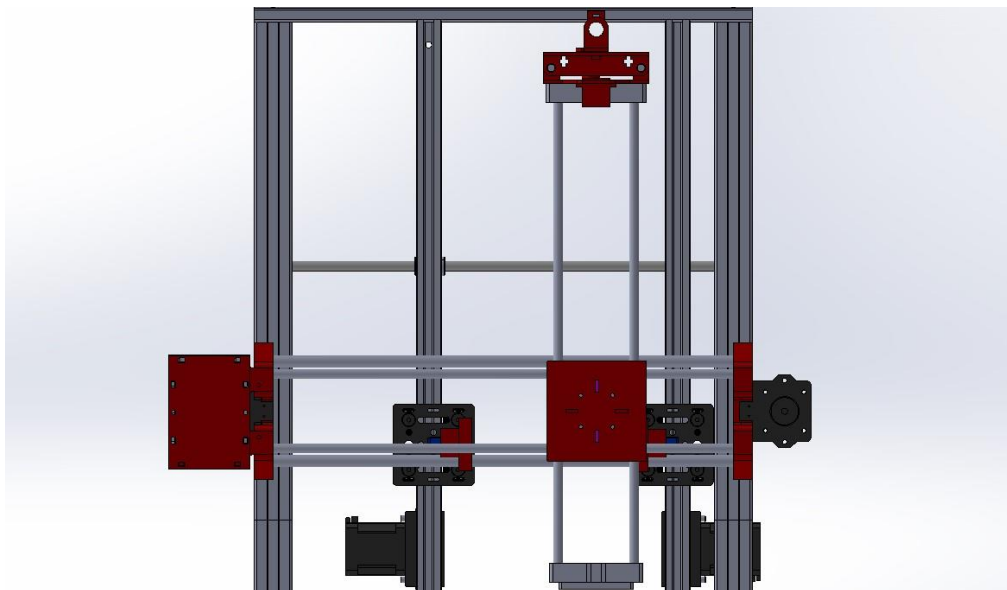


Figure 3.19:Pen Plotter Station

3.3.7 Standard Components and Fasteners

All mechanical assemblies utilize standard off-the-shelf components:

Linear Motion Components:

- Linear bearings: LM8UU (8mm bore, 15mm OD, 24mm length)
- Linear shafts: 8mm diameter, hardened steel
- Quantity: 12 units (4 per CNC station $\times$ 3 axes)

Fasteners:

- M3 screws: General assembly (6mm, 8mm, 10mm, 12mm lengths)
- M4 screws: Motor mounting (10mm, 16mm lengths)
- M5 screws: Frame connections (12mm, 20mm lengths)
- M8 screws: Base mounting (25mm length)
- Nuts: Nylock nuts for vibration resistance
- Washers: Flat washers and spring washers

Brackets and Connectors:

- Aluminum profile corner brackets (90°)
- T-slot nuts (M3, M4, M5)
- 3D-printed custom brackets (PLA/PETG)
- Motor mounting plates (3mm aluminum)

Table 3-2: Bill Of Material

Num	Part Name	Specification	Material	Quantity
1	Aluminum Profile 3030	30×30 mm, T-slot groove	Aluminum 6063-T5	~12 m
2	Aluminum Profile 2020	20×20 mm, T-slot groove	Aluminum 6063-T5	~8 m
3	Linear Bearing	LM8UU (8×15×24 mm)	Chrome Steel	12 pcs
4	Linear Shaft	Ø8 mm, hardened & ground	Carbon Steel	~15 m
5	Lead Screw	T8 (Ø8 mm, 2 mm pitch)	Carbon Steel	3 pcs
6	Anti-Backlash Nut	T8 compatible, double-nut type	Brass	3 pcs
7	Timing Belt	GT2 profile, 6 mm width, 2 mm pitch	Rubber/Fiberglass	~8 m
8	GT2 Pulley	16-tooth, 5 mm bore, 6 mm width	Aluminum	8 pcs
9	Socket Head Screws (M3)	6–12 mm length	Stainless Steel 304	~50 pcs
10	Socket Head Screws (M4)	10–20 mm length	Stainless Steel 304	~30 pcs
11	Socket Head Screws (M5)	12–25 mm length	Stainless Steel 304	~20 pcs
12	Nylock Nuts	M3, M4, M5	Stainless Steel 304	~60 pcs
13	T-Slot Nuts	M3, M4, M5	Carbon Steel	~40 pcs
14	Corner Brackets	90° L-bracket, reinforced	Aluminum Alloy	24 pcs
15	Motor Mount Plates	Custom cut, 3 mm thick	Aluminum 6061	6 pcs
16	3D Printed Structural Parts	U-gripper, latches, base mounts	PETG Filament	12 sets
17	3D Printed Tooling Parts	Pen holder, spray gun adapter	PLA / PETG	4 sets
18	Flexible Coupling	Jaw/beam type, 5×8 mm bore	Aluminum/TPU	3 pcs
19	Base Plate	Custom size, 3–5 mm thickness	Aluminum Sheet	2 pcs
20	Cable Management	Nylon clips & zip ties	Nylon PA66	~20 pcs

3.4 CONTROL SYSTEM DESIGN

3.5 SAFETY CONSIDERATIONS

CHAPTER 4: Implementation and Testing

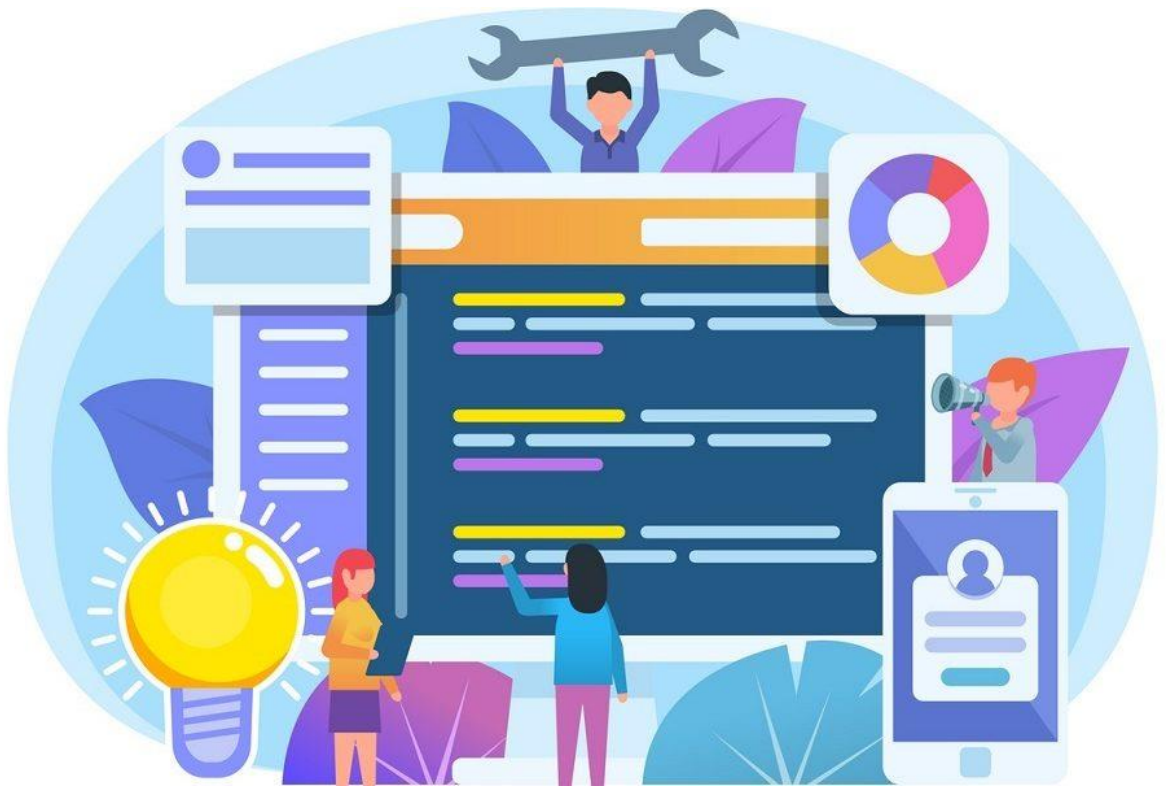


Figure 4.1:Implementation and Testing

4.1 HARDWARE IMPLEMENTATION

This section details the physical assembly and integration of all mechanical, electrical, and electromechanical components that constitute the automated PCB production line. The hardware implementation was carried out in a systematic manner, beginning with the structural frame assembly, followed by the installation of conveyor systems, actuation mechanisms, sensor networks, and final electrical wiring.

4.1.1 System Overview and Workstation Layout

The automated PCB production line comprises two main workstations, each constructed using modular aluminum extrusion profiles (6063-T5 alloy). The first workstation houses the spray coating chamber and CNC drilling station, while the second workstation contains the CNC marking (pen plotter) station and manual assembly area. A comprehensive control panel integrates all power supplies, microcontrollers, and operator interface components.

4.1.2 Control Panel and Power Distribution System

The electrical control system is housed within a dedicated control panel mounted beneath the first workstation. This panel integrates all power supply units, microcontrollers, relay modules, and operator interface components necessary for system operation.

4.1.2.1 Power Supply Configuration

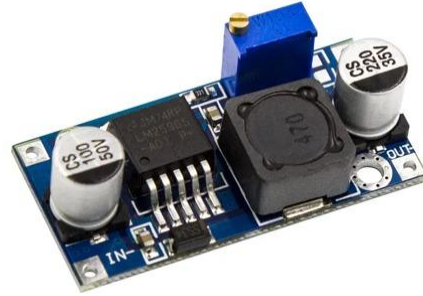
Component	Specification	Function
Main Power Supply	12 V DC, 30 A (360 W)	Primary power source for stepper motors, servo motors, conveyor motors, and relay coils
DC-DC Buck Converter	LM2596S (12V $\rightarrow$ 5V, 3A)	Provides regulated 5 V DC for Arduino boards, sensor logic, and servo control signals
Toggle Switch	10 A, ON/OFF	Master power switch for the entire system

- The 30 A power supply was selected based on total system power calculations:

$$P_{total} = (V_{motor} \times I_{total}) + (V_{logic} \times I_{logic})$$



Power Supply 12V 30A



DC-DC Buck Converter LM2596

Component	Voltage (V)	Current (A)	Power _{quantity} (W)
Stepper Motors (×6)	12	1.7	$12 \times 1.7 \times 6 = 122.4$
Servo Motors (×5)	5	0.9	$5 \times 0.9 \times 5 = 22.5$
DC Spindle Motor (×1)	12	5.0	$12 \times 5.0 = 60$
Conveyor Motors (×4)	12	1.5	$12 \times 1.5 \times 4 = 72$
Subtotal	276.9 W		
Safety Margin (20%)	$276.9 \times 0.20 = 55.38 \text{ W}$		
Total	$276.9 + 55.38 = 332.28 \text{ W}$		

Safety Margin (20%):

$$P_{margin} = 276.9 \times 0.20 = 55.38 \text{ W}$$

$$P_{total} = 122.4 + 22.5 + 60 + 72 = 276.9 \text{ W}$$

$$P_{total \text{ with margin}} = 276.9 \times 1.20 = 332.3 \text{ W}$$

$$I_{total} = \frac{P_{total}}{V} = \frac{332.3}{12} \approx 27.7 \text{ A}$$

- The **30 A power supply** provides adequate capacity with a safety margin of approximately 8%.

Total Current:

$$I_{total} = \frac{P_{total}}{V} = \frac{332.3}{12} \approx 27.7 \text{ A}$$

Safety Margin on Power Supply:

$$Safety_{margin} = \frac{30 - 27.7}{30} \times 100 = \frac{2.3}{30} \times 100 \approx 7.67\% \approx 8\%$$

4.1.2.2 Operator Interface and Status Indicator

Component	Function
Toggle Switch (ON/OFF)	Master power control
Green Indicator Lamp	Illuminates when system is ON
Red Indicator Lamp	Illuminates when system is OFF

The dual-lamp configuration provides immediate visual feedback to operators regarding the system's power state, enhancing safety during maintenance and setup operations.



Inductor 220V RED



Selector Switch (Toggle)

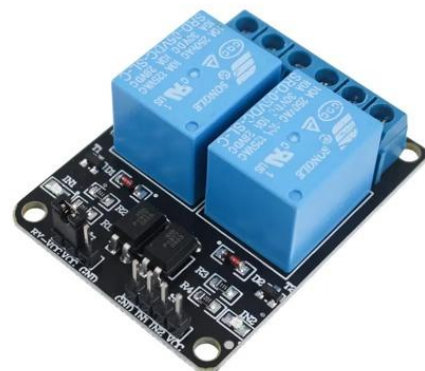


Inductor 220V GREEN

4.1.2.3 Relay Module Integration

A 2-Channel Relay Module, 5V DC, 10A are installed within the control panel:

Relay	Function	Controlled Device
Relay 1 (CH1)	Spray system actuation	Air compressor / Solenoid valve
Relay 2 (CH2)	CNC spindle control	DC spindle motor (775, 9000 RPM)



Relay Module 2 Channel – 5VDC

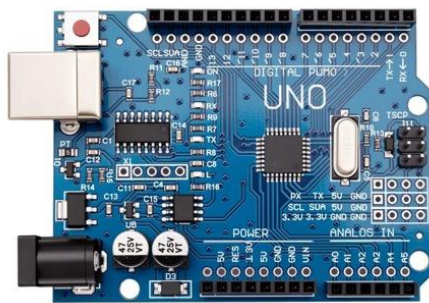
Both relay modules are optically isolated and include flyback diode protection to suppress inductive voltage spikes generated by the inductive loads (solenoids and motors).

4.1.2.4 Microcontroller Architecture

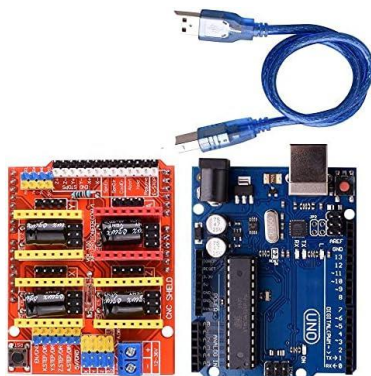
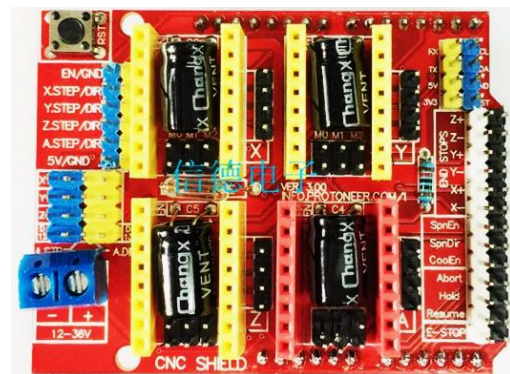
The system utilizes a distributed control architecture comprising **three Arduino Uno with shield** boards and one **ESP32 (ESP-WROOM-32)** module:

Controller	Model	Purpose	Key Peripherals
Arduino Uno #1	ATmega328P	Spray chamber control	NEMA 17 (X-axis), SG90 (nozzle), conveyor motors
Arduino Uno #2	ATmega328P	CNC drilling station (Workstation 1)	NEMA 17 (X, Y, Z), DC spindle, limit switches
Arduino Uno #3	ATmega328P	CNC marking station (Workstation 2)	NEMA 17 (X, Y), SG90 (Z-axis/pen lift)
ESP32 Module	ESP-WROOM-32	Sensor interface & status monitoring	Red/Green clamping buttons, status LEDs

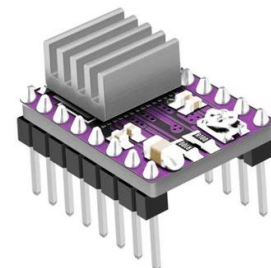
Each Arduino board is equipped with a **CNC Shield V3** providing dedicated sockets for **A4988 stepper motor drivers**.



Arduino Uno R3



ESP32



DRV8825

4.1.3 Workstation 1: Spray Chamber and CNC Drilling

4.1.3.1 Conveyor System (Workstation 1)

Component	Specification	Function
Stepper Motors	NEMA 17 (×2)	Drive GT2 timing belt conveyor
Belt Type	GT2, 6 mm width, 2 mm pitch	Positive engagement transport
Pulley	16-tooth, 10.2 mm pitch diameter	Drive and idler pulleys

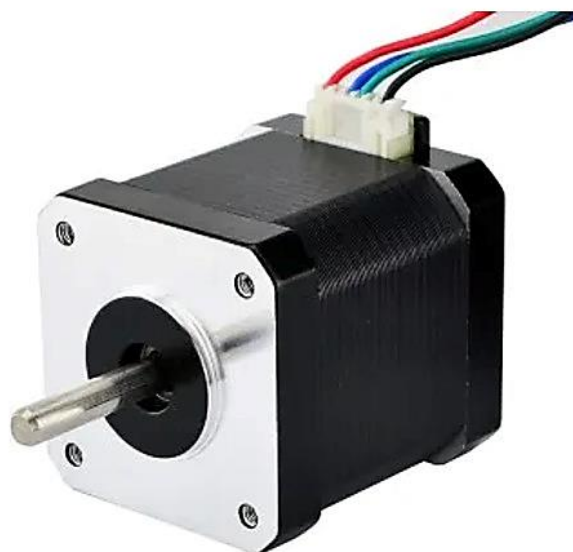
Conveyor Speed Calculation:

$$v = \frac{\pi \times D \times N}{60}$$

Where:

- D = Pulley diameter = 10.2 mm = 0.0102 m
- N = Motor speed = 300 RPM (typical)

$$v = \frac{\pi \times 0.0102 \times 300}{60} = 0.160 \text{ m/s} = 160 \text{ mm/s}$$



Stepper Motor (NEMA17)

4.1.3.2 PCB Clamping Mechanism (Workstation 1)

Component	Quantity	Specification	Function
Servo Motors	2	SG90 (Blue, 180° rotation, 1.2–1.8 kg·cm)	Open/close gripper jaws
Red Push Button	1	Push button (NO)	Release PCB (OPEN)
Green Push Button	1	Push button (NO)	Clamp PCB (CLOSE)

Clamping Force Calculation:

$$F_{clamp} = \frac{T_{servo} \times \eta}{r}$$

Where:

- $T_{servo} = 1.8 \text{ kg}\cdot\text{cm} = 0.176 \text{ N}\cdot\text{m}$ (SG90 torque)
- $\eta = 0.85$ (mechanical efficiency of linkage)
- $r = 5 \text{ mm} = 0.005 \text{ m}$ (lever arm length)



Servo Motor SG90

$$F_{clamp} = \frac{0.176 \times 0.85}{0.005} = 29.9 \text{ N} \approx 3.0 \text{ Kg}$$

Operation Sequence:

1. When a PCB is placed onto the carriers, the red limit switch triggers a stop command to the stepper motors.
2. The SG90 servos actuate the grippers to clamp the PCB.
3. Upon clamping confirmation from the green limit switch, the conveyor restarts.



Pushbutton RED



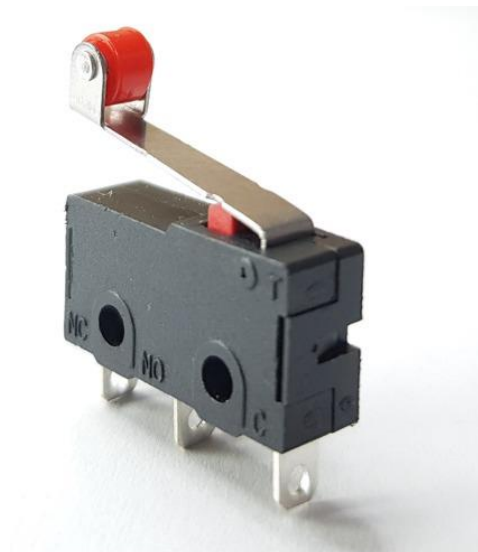
Pushbutton GREEN

4.1.3.3 Limit Switch Network (Workstation 1)

Switch ID	Position	Function
LS1	After spray chamber	Signals completion of spraying
LS2	Before CNC drilling station	Triggers homing and drilling cycle
LS3	Before flipping mechanism	Positions carrier for flipping



Limit Switch With Roller Wheel (MS.3)



Limit Switch With Roller Wheel (MS.2)

4.1.3.4 Spray Chamber Assembly

Component	Specification	Function
Stepper Motor (X-axis)	NEMA 17	Moves spray nozzle horizontally
Servo Motor (Nozzle)	SG90 (Blue, 180° rotation, 1.2–1.8 kg·cm)	Controls nozzle angle and on/off
Spray Gun	Airbrush type, 0.3 mm nozzle	Applies insulation coating
Compressor	2–4 bar	Atomizes coating material

Coating Thickness Estimation:

$$t = \frac{Q \times N_p}{v \times W}$$



Servo Motor SG90

Where:

- Q = Flow rate = 10 mL/min = 167 mm³/s
- N_p = Number of passes = 2
- v = Traverse speed = 50 mm/s
- W = Spray width = 20 mm

$$t = \frac{167 \times 2}{50 \times 20} = 0.334 \text{ mm} = 334 \mu\text{m}$$

4.1.3.5 CNC Drilling Station

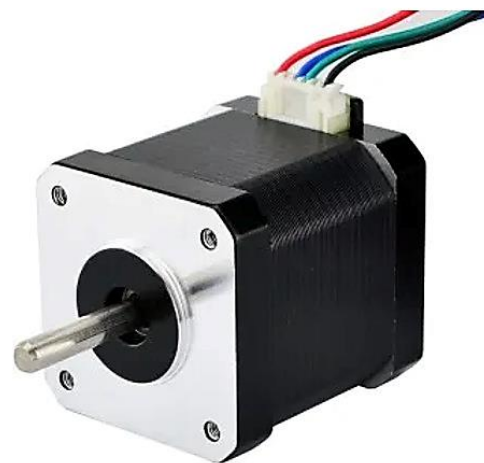
Component	Specification	Function
Stepper Motors (X, Y)	NEMA 17 (×2)	Move XY table
Stepper Motor (Z)	NEMA 17 (×1)	Moves spindle vertically
DC Spindle Motor	Model 775, 9000 RPM, 12–36 V	Drilling and routing
Lead Screws	T8, 2 mm pitch, anti-backlash nuts	Precision positioning
Linear Bearings	LM8UU (×12, 8 mm bore)	Smooth linear motion
Linear Shafts	8 mm hardened steel	Guide rails

Positional Resolution:

$$Resolution = \frac{Lead}{Steps/Rev \times Microsteps}$$

Where:

- Lead = 2 mm/rev
- Steps/Rev = 200 (NEMA 17)
- Microsteps = 16 (A4988 configuration)



Stepper Motor (NEMA17)

$$Resolution = \frac{2}{200 \times 16} = 0.000625 \text{ mm/step} = 0.625 \text{ } \mu\text{m/step}$$

4.1.3.6 Spindle Motor Assembly (Model 775)

Parameter	Value
Motor Type	Brushed DC Motor
Model	775
Rated Speed	9000 RPM (no-load at 12 V)
Operating Voltage	12 – 36 V DC
Speed Range	3000 – 9000 RPM (voltage dependent)
Current Draw	5–10 A (under load)
Torque	1.2 kg·cm ($\approx 0.117 \text{ N}\cdot\text{m}$)
Shaft Diameter	5 mm
Power Rating	60–150 W
Weight	0.35 kg
Dimensions	42 × 66 mm
Mounting	Custom aluminum bracket with hose clamps



Speed-Voltage Relationship:

$$\text{Speed (RPM)} = \frac{V_{\text{applied}}}{12} \times 9000$$

Where:

- $V_{\text{applied}} = 0 - 36 \text{ V}$ (regulated via PWM from Arduino)
- At 12 V: **9000 RPM**
- At 24 V: **18000 RPM** (theoretical)

Note: The system power supply is limited to 12 V, providing a nominal speed of 9000 RPM at full duty cycle. This speed is sufficient for PCB drilling and routing applications using tungsten carbide tools.

4.1.4 PCB Flipping Mechanism

Component	Specification	Function
Servo Motor	MG995 (Black, 180°, 10 kg·cm, Metal Gears)	Rotates gripper 180°
Gripper Structure	3D-printed PETG, U-shaped	Holds PCB during flipping
Clamping Servos	SG90 (×2, Blue, 180°)	Secure PCB in gripper

Torque Requirement:

$$T_{required} = m \times g \times r = 0.15 \times 9.81 \times 0.05 = 0.074 \text{ N} \cdot \text{m} = 0.75 \text{ kg} \cdot \text{cm}$$

Safety Factor:

$$SF = \frac{10}{0.75} = 13.3$$

Flipping Cycle Time:

$$t_{flip} = \frac{180}{60} \times 0.17 = 0.51 \text{ s} \Rightarrow t_{total} \approx 1.5 \text{ s} < 3 \text{ s}$$



4.1.5 Workstation 2: CNC Marking and Assembly

4.1.5.1 Conveyor System (Workstation 2)

Component	Specification	Function
Stepper Motors	NEMA 17 (×2)	Drive GT2 timing belt conveyor
Limit Switch #1	Mechanical NO	Detects PCB arrival; stops conveyor and initiates marking sequence
Limit Switch #2	Mechanical NO	Detects completed marking; stops conveyor for manual assembly

4.1.5.2 PCB Clamping Mechanism (Workstation 2)

Component	Quantity	Specification	Function
Servo Motors	2	SG90 (Blue, 180°)	Open/close gripper jaws
Red Push Button	1	Push button (NO)	Release PCB (OPEN)
Green Push Button	1	Push button (NO)	Clamp PCB (CLOSE)

Operation Sequence:

1. The PCB arrives at the marking station, triggered by Limit Switch #1.
2. The SG90 servo lowers the pen to the PCB surface.
3. The NEMA 17 motors move the pen along the programmed G-code path.
4. Upon completion, the SG90 servo lifts the pen.
5. The conveyor resumes and the PCB moves toward Limit Switch #2 for manual assembly.

Tool Path Generation:

The marking paths are generated from the PCB design data (Gerber files) using a Python script that extracts component reference designators (R1, C5, U3, etc.) and converts them to G-code commands. The system supports:

- **Text marking:** Component designators and polarity indicators
- **Outline marking:** Component body outlines for placement guidance
- **Alignment marks:** Fiducial markers for assembly verification

4.1.5.3 CNC Marking Station (Pen Plotter)

Component	Specification	Function
Stepper Motors	NEMA 17 (×2)	Drive X and Y axes
Servo Motor (Z-axis)	SG90 (Blue, 180°)	Pen lift mechanism
Pen Holder	3D-printed clamp	Holds 8–12 mm pens
Travel Range	X: 300 mm, Y: 200 mm	Covers PCB sizes

Resolution Calculation:

For GT2 belt with 16-tooth pulley:

$$\begin{aligned}\text{Travel/Rev} &= \pi \times D = \pi \times 10.2 = 32.0 \text{ mm} \\ \text{Resolution} &= \frac{32.0}{200 \times 16} = 0.01 \text{ mm/step} = 10 \text{ }\mu\text{m/step}\end{aligned}$$

Compatibility with GRBL:

The marking station uses the same **GRBL v1.1h** firmware as the drilling station, with the following customizations:

- **X and Y axes:** Configured for belt drive (160 steps/mm)
- **Z-axis:** Replaced with SG90 servo (pen lift) instead of stepper motor
- **Spindle control:** Disabled (not used for pen plotting)

4.1.6 Wiring and Cable Specifications

Cable Type	Application	Specification
Power Cable	12V/5V distribution	18 AWG, twisted pair
Signal Cable	Limit switches, sensors	22 AWG, shielded
Motor Cable	Stepper motors (NEMA 17)	24 AWG, 4-core
Servo Cable	SG90 (grippers, nozzle, pen) & MG995 (flipping)	26 AWG, 3-wire (VCC, GND, PWM)
DC Motor Cable	Spindle motor (Model 775, 9000 RPM, 12–36V)	18 AWG, 2-wire (VCC, GND)
USB Cable	Arduino communication	USB-B to USB-A, shielded

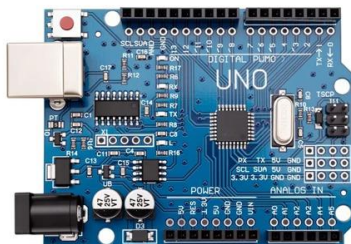
4.1.6.1 Control Panel Wiring

The control panel houses the power supply, buck converter, relay modules, and ESP32 module. The wiring connections are as follows:

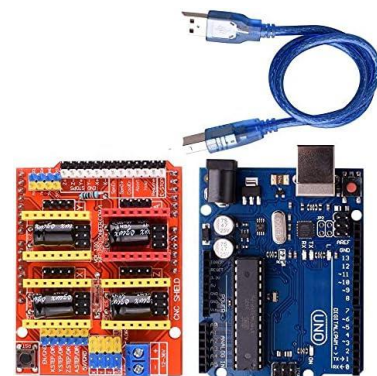
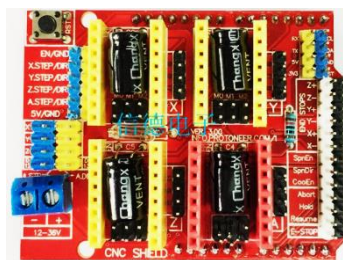
From	To	Cable Type	Specification
12V Power Supply	DC-DC Buck Converter (IN+)	Power Cable	18 AWG
12V Power Supply	Relay Modules (COM)	Power Cable	18 AWG
12V Power Supply	CNC Shields (VMOT)	Power Cable	18 AWG
12V Power Supply	Toggle Switch (IN)	Power Cable	18 AWG
Buck Converter (OUT+)	Arduino 5V Pins	Power Cable	22 AWG
Buck Converter (GND)	Common Ground Bus	Power Cable	22 AWG
Relay 1 (NO)	Spray Solenoid Valve	Power Cable	18 AWG
Relay 2 (NO)	DC Spindle Motor (775)	Power Cable	18 AWG
Relay 1 (IN)	Arduino #1 (D10)	Signal Cable	22 AWG, shielded
Relay 2 (IN)	Arduino #2 (D10)	Signal Cable	22 AWG, shielded
ESP32 (GPIO0)	Red Push Button	Signal Cable	22 AWG, shielded
ESP32 (GPIO2)	Green Push Button	Signal Cable	22 AWG, shielded
ESP32 (TX/RX)	Arduino UART	Signal Cable	22 AWG, shielded
Arduino USB	Host PC	USB Cable	USB-B to USB-A, shielded

4.2 (TABLE) ARDUINO PIN ASSIGNMENT SUMMARY

Arduino	Pin	Function
#1 (Spray)	D4	Step X (Spray)
	D5	Dir X (Spray)
	D6	Step Conveyor W1
	D7	Dir Conveyor W1
	D9	SG90 (Nozzle)
	D10	Relay 1 (Spray)
#2 (Drill)	D4	Step X (Drill)
	D5	Dir X (Drill)
	D6	Step Y (Drill)
	D7	Dir Y (Drill)
	D8	Step Z (Drill)
	D9	Dir Z (Drill)
	D10	Relay 2 (Spindle)
	D11	PWM Speed
#3 (Mark)	D4	Step X (Mark)
	D5	Dir X (Mark)
	D6	Step Y (Mark)
	D7	Dir Y (Mark)
	D9	SG90 (Pen)



Arduino Uno R3



4.3 SOFTWARE IMPLEMENTATION

4.4 SYSTEM INTEGRATION

4.5 TESTING PROCEDURES

4.6 RESULTS AND ANALYSIS

4.7 CHALLENGES AND SOLUTIONS

4.8 SUMMARY

CHAPTER 5: Conclusion and Future Work



Figure 5.1: Conclusion and Future Work

5.1 CONCLUSION

This graduation project successfully designed, developed, and validated an automated PCB production line prototype integrating mechanical, electrical, and embedded control systems. The system automates critical manufacturing stages including PCB feeding, CNC-based drilling and routing, board flipping for double-sided processing, precision marking of component locations, and guided assembly preparation. An Arduino-based control architecture coordinates servo motors, stepper drives, limit sensors, and G-code execution to ensure synchronized operation across all stations.

Functional testing demonstrated consistent repeatability, acceptable positional accuracy, and reliable process sequencing, validating the feasibility of low-cost embedded automation for small-scale PCB fabrication. The project highlights the practical application of mechatronics principles in industrial automation, providing a scalable framework that bridges academic design with real-world manufacturing requirements. By reducing manual intervention, improving process consistency, and establishing a modular workflow, the developed prototype achieves its core objectives and serves as a functional foundation for further industrial optimization

5.2 FUTURE WORK

While the current prototype successfully demonstrates the core automation workflow, several enhancements can be implemented to elevate the system toward industrial-grade performance:

5.2.1 Control System Migration: Transitioning from Arduino-based control to industrial PLCs or real-time embedded controllers (e.g., STM32/ESP32 with RTOS) will improve deterministic timing, fault tolerance, and scalability for continuous production environments.

5.2.2 Machine Vision Integration: Incorporating camera-based inspection systems for automated optical inspection (AOI) and fiducial alignment will enable real-time defect detection, layer registration verification, and closed-loop positional correction during drilling and marking.

5.2.3 IoT and Industry 4.0 Connectivity: Implementing networked communication protocols (MQTT/OPC UA) will allow remote monitoring, production data logging, predictive maintenance analytics, and integration with Manufacturing Execution Systems (MES).

5.2.4 Adding advanced stages such as tin and flux tanks with mechanisms that lower the board into those tanks and then raise it up to create a perfect product ready for immediate testing.

5.2.5 • Enhanced HMI and Safety Systems: Developing a full SCADA interface with recipe management, user authentication, and emergency stop interlocks will improve operator control, data traceability, and compliance with industrial safety standards (e.g., IEC 60204-1).

These proposed developments will transform the current academic prototype into a robust, industry-ready manufacturing solution while maintaining the project's core philosophy of cost-effective, modular automation.

5.3 REFERENCES

- 5.3.1 C. F. Coombs Jr. and H. Holden, Eds., Printed Circuits Handbook, 7th ed. New York, NY, USA: McGraw-Hill Education, 2016.**
- 5.3.2 Mechatronics: Electronic Control Systems in Mechanical and Electrical Engineering W. Bolton, Mechatronics: Electronic Control Systems in Mechanical and Electrical Engineering, 7th ed. Harlow, U.K.: Pearson Education, 2019.**
- 5.3.3 A. Overby, CNC Machining Handbook: Building, Programming, and Implementation. New York, NY, USA: McGraw-Hill Education, 2010.**
- 5.3.4 International Electrotechnical Commission, Safety of Machinery – Electrical Equipment of Machines – Part 1: General Requirements, IEC Standard 60204-1, 2016.**
- 5.3.5 IPC, Generic Standard on Printed Board Design, IPC-2221, Bannockburn, IL, USA, 2012.**
- 5.3.6 IPC, Qualification and Performance Specification for Rigid Printed Boards, IPC-6012, Bannockburn, IL, USA, 2015.**
- 5.3.7 IPC, Requirements for Soldered Electrical and Electronic Assemblies, IPC-J-STD-001, Bannockburn, IL, USA, 2020.**
- 5.3.8 Arduino Language Reference, Arduino Documentation, 2026. [Online]. Available: <https://docs.arduino.cc/language-reference/>. [Accessed: May 20, 2026].**
- 5.3.9 K. H. Kim and S. J. Park, “Design and control of a modular CNC drilling system for PCB prototyping,” International Journal of Advanced Manufacturing Technology, vol. 98, no. 5–8, pp. 1123–1135, Oct. 2018.**
- 5.3.10 A. Hassan and M. Ali, “Arduino-based automation for small-scale electronic manufacturing lines,” in Proc. IEEE Int. Conf. Mechatronics and Automation, 2020, pp. 455–460.**
- 5.3.11 R. Gupta and L. Chen, “Closed-loop servo control for precision positioning in automated assembly systems,” Journal of Mechatronics Engineering, vol. 12, no. 2, pp. 67–79, Mar. 2021.**
- 5.3.12 M. Martinez and K. Wong, “Development of a low-cost automated dispensing system for small-batch PCB production,” Journal of Electronics Manufacturing, vol. 32, no. 4, pp. 245–258, Jun. 2023.**

5.3.13 J. Lee, H. Park, and S. Kim, “Investigation of spray coating parameters for conformal coating on printed circuit boards,” Surface and Coatings Technology, vol. 435, pp. 128–142, Apr. 2022.

5.3.14 International Electrotechnical Commission, Electromagnetic Compatibility (EMC) – Part 3-2: Limits – Limits for Harmonic Current Emissions, IEC Standard 61000-3-2, 2018.

5.3.15 E. E. Rebecca, “Alternating current fed power supply,” U.S. Patent 7 897 777, Nov. 3, 2011.

5.3.16 S. Mack, “Automated control systems in modern manufacturing,” M.A. thesis, University of Calgary, Calgary, AB, Canada, 2022.